POWER ELECTRONICS (PE) — MODULE 2

The Definitive Master Study Guide, Waveform Atlas & Derivation Handbook

Course: B.Tech S5 Electrical & Electronics Engineering (EEE) | College of Engineering Trivandrum (CET)

Integrated References: Dr. P. S. Bimbhra, Ned Mohan, Daniel W. Hart, L. Umanand & Official Course Lecture Waveforms

Contents

Module 2 Syllabus, Architecture & Learning Roadmap

According to the official APJ Abdul Kalam Technological University / CET syllabus, Module 2 carries substantial weightage in both Continuous Internal Evaluation (CIE) and End Semester Examination (ESE). It covers three major areas:

1. **Single-Phase Controlled Rectifiers (5 Hours):** Fully controlled and half-controlled (semi-converter) rectifiers with RL and RLE loads; rectifier and inverter modes of operation; continuous and discontinuous conduction waveforms; output voltage, input line current, real power, power factor (PF), and total harmonic distortion (THD); effect of source inductance (L_s) with commutation overlap angle μ.
2. **Three-Phase Controlled Rectifiers (3 Hours):** 6-pulse fully controlled bridge and half-controlled bridge converter with RLE load (continuous conduction, ripple-free current); conduction sequence, firing intervals, waveforms, and output voltage equations; source inductance effect.
3. **DC-DC Switching Regulators (4 Hours):** Buck, Boost, and Buck-Boost converters; continuous conduction mode (CCM) operation and waveforms; Inductor Volt-Second Balance and Capacitor Charge Balance; design of filter inductance (L_{min}) and capacitance (C_{min}); qualitative and quantitative effects of non-idealities (inductor resistance r_L, capacitor ESR r_C); selection criteria for power semiconductor switches (MOSFET, IGBT, Schottky diode).

Bloom's Knowledge Level K4 (Analyse)

In your examinations (Part A 3-mark questions and Part B 9-mark questions), you are expected to:

- Sketch clear, aligned waveforms showing supply voltage, gate pulses, load voltage, switch voltage, and input current.
- Derive mathematical expressions from first principles using calculus and integral definitions.
- Compute exact numerical values for output voltage, RMS quantities, displacement factor, distortion factor, power factor, and filter components.

Foundations of Phase Control & Commutation Mechanics

What is Phase-Controlled Rectification?

An uncontrolled diode conducts whenever its anode-to-cathode voltage is positive ($v_{AK} > 0$). In contrast, a **Silicon Controlled Rectifier (SCR / Thyristor)** blocks forward voltage until a triggering gate pulse i_G is applied. By delaying this gate trigger pulse by an angle α with respect to the AC voltage zero-crossing:

Average output DC voltage V_o can be smoothly and steplessly controlled from maximum to zero (or negative).

Natural (Line) Commutation vs. Forced Commutation

- **Natural (Line) Commutation:** In AC-fed circuits, the AC line voltage reverses polarity every half cycle. When the supply voltage reverses, a negative voltage is placed across the conducting SCR, naturally reducing its current below holding current (I_H) and providing reverse bias for turn-off time t_q. No auxiliary components are needed.
- **Forced Commutation:** Required when the supply is DC (classical choppers/inverters), necessitating auxiliary capacitors/inductors to force SCR current to zero.

Key Angular Definitions

- **Firing Angle (Delay Angle) α:** The angle measured from the instant the thyristor would naturally conduct if it were an ordinary diode (the *natural commutation point*) to the instant the gate pulse is injected.
- **Extinction Angle β:** The instant $\omega t = \beta$ at which the thyristor anode current drops to zero.
- **Conduction Angle γ:** The angular duration over which the device conducts: $\gamma = \beta - \alpha$.

Converter Quadrants of Operation

Thyristor Current Unidirectionality

Because an SCR is a single-direction semiconductor switch, the DC load current I_o can **never reverse**:

$$I_o \geq 0 \quad \text{(Strictly unidirectional)}$$

However, the average DC output voltage V_o **can reverse** polarity depending on α and circuit topology!

- **Quadrant I ($V_o > 0, I_o > 0$):** Rectification Mode. Power flows from AC source to DC load.
- **Quadrant IV ($V_o < 0, I_o > 0$):** Inverter Mode. Power flows from DC active source back into the AC utility grid.
- **Semi-Converter:** Operates in **Quadrant I only** (One-quadrant converter).
- **Full Converter:** Operates in **Quadrants I and IV** (Two-quadrant converter).

Single-Phase Controlled Rectifiers (Deep Dive)

Single-Phase Half-Wave Rectifiers: Complete Load Analysis

Resistive Load (R)

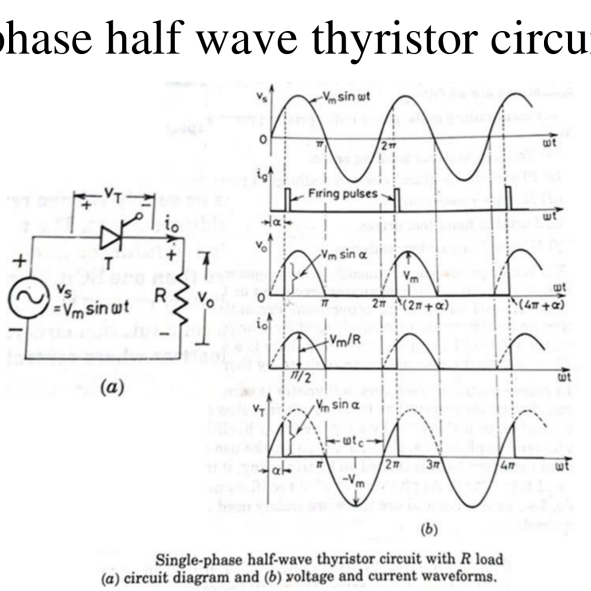

Figure 1: Single-phase half-wave controlled rectifier with resistive load: Circuit and complete waveforms showing firing delay α, load voltage v_o, and device voltage v_T.

How it works: The AC supply voltage $v_s = V_m \sin \omega t$ starts its positive half cycle at $\omega t = 0$. Even though the SCR becomes forward-biased at this instant, it *does not conduct* — it waits in the blocking state until a gate pulse arrives at $\omega t = \alpha$. This delay angle α is our control variable.

Once triggered at α, the SCR turns ON and the full supply voltage appears across the resistive load: $v_o = v_s = V_m \sin \omega t$. With a purely resistive load, the current is simply $i_o = v_o/R$, which exactly follows the voltage shape. At $\omega t = \pi$, the supply voltage crosses zero and tries to go negative. Since a resistive load cannot sustain current flow against a negative voltage, the current drops to zero and the SCR naturally turns OFF.

Result: The load sees voltage only during $\alpha \leq \omega t \leq \pi$ in each cycle, and zero voltage elsewhere.

Average (DC) output voltage is the total area under the v_o curve divided by the full period 2π:

$$V_o = \frac{1}{2\pi} \int_{\alpha}^{\pi} V_m \sin \omega t \, d(\omega t) = \frac{V_m}{2\pi} [-\cos \omega t]_{\alpha}^{\pi} = \frac{V_m}{2\pi} (1 + \cos \alpha) \tag{1}$$

RMS output voltage (needed for power and ripple calculations):

$$V_{o,\text{rms}} = \frac{V_m}{2} \sqrt{\frac{1}{\pi} \left[(\pi - \alpha) + \frac{1}{2} \sin(2\alpha) \right]} \tag{2}$$

Notice that at $\alpha = 0$, $V_o = V_m/\pi$ (same as an uncontrolled diode rectifier), and at $\alpha = \pi$, $V_o = 0$ (SCR never conducts).

Inductive Load (RL) — Why Negative Voltage Occurs

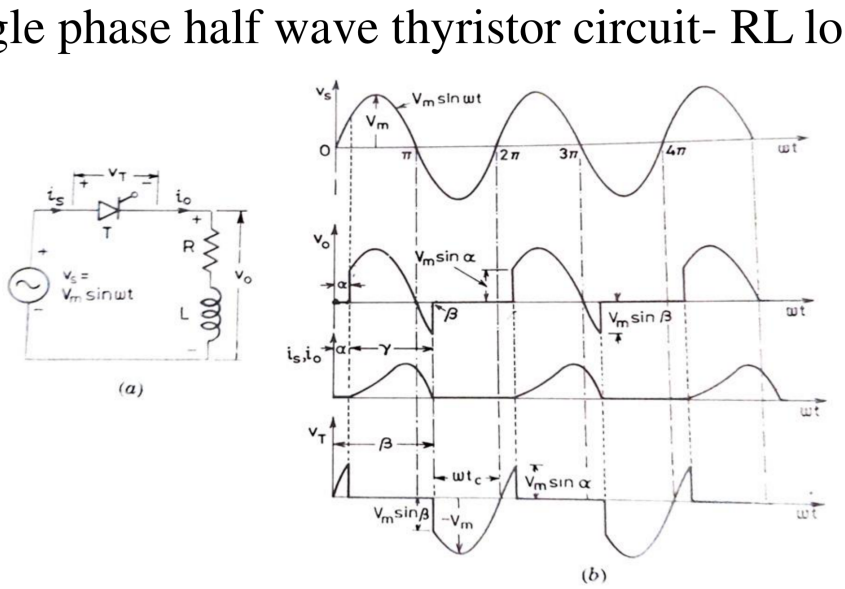

Figure 2: Single-phase half-wave circuit with RL load: Waveforms showing conduction extending into the negative half-cycle up to extinction angle $\beta > \pi$ due to inductor induced EMF.

When an inductor is present in series with R:

- During $\alpha \leq \omega t \leq \pi$, the inductor stores energy in its magnetic field: $E_L = \frac{1}{2}Li^2$.
- At $\omega t = \pi$, supply voltage v_s crosses zero and goes negative.
- The collapsing magnetic field of the inductor induces a voltage:

$$v_L = L\frac{di}{dt} < 0$$

  which acts as a source in series with the AC line, keeping the thyristor anode positive with respect to cathode!
- Consequently, the thyristor continues to conduct into the negative half-cycle until all magnetic energy is discharged at extinction angle $\beta > \pi$.
- The output voltage is negative during $\pi \leq \omega t \leq \beta$:

$$V_o = \frac{V_m}{2\pi}(\cos\alpha - \cos\beta)$$

General Current Solution for Half-Wave RL Load

Applying KVL to the circuit during conduction ($\alpha \leq \omega t \leq \beta$):

$$V_m \sin\omega t = R\,i + L\frac{di}{d(\omega t)}\,\omega$$

Let $Z = \sqrt{R^2 + (\omega L)^2}$ be the load impedance and $\phi = \tan^{-1}\left(\frac{\omega L}{R}\right)$ the impedance angle. The complete solution (particular + complementary) is:

$$i(\omega t) = \frac{V_m}{Z}\left[\sin(\omega t - \phi) - \sin(\alpha - \phi)\, e^{-\frac{R}{\omega L}(\omega t - \alpha)}\right], \quad \alpha \leq \omega t \leq \beta$$

- The first term $\frac{V_m}{Z}\sin(\omega t - \phi)$ is the **steady-state (forced) response**.

- The second term is the **transient (natural) response** that decays exponentially with time constant $\tau = L/R$.
- The extinction angle β is found by setting $i(\beta) = 0$ (requires numerical/graphical solution in general).
- **Condition for continuous conduction:** If $\alpha \leq \phi$, the thyristor is triggered before the current from the previous cycle has decayed to zero, leading to continuous conduction mode.

Action of Freewheeling Diode (D_{fw})

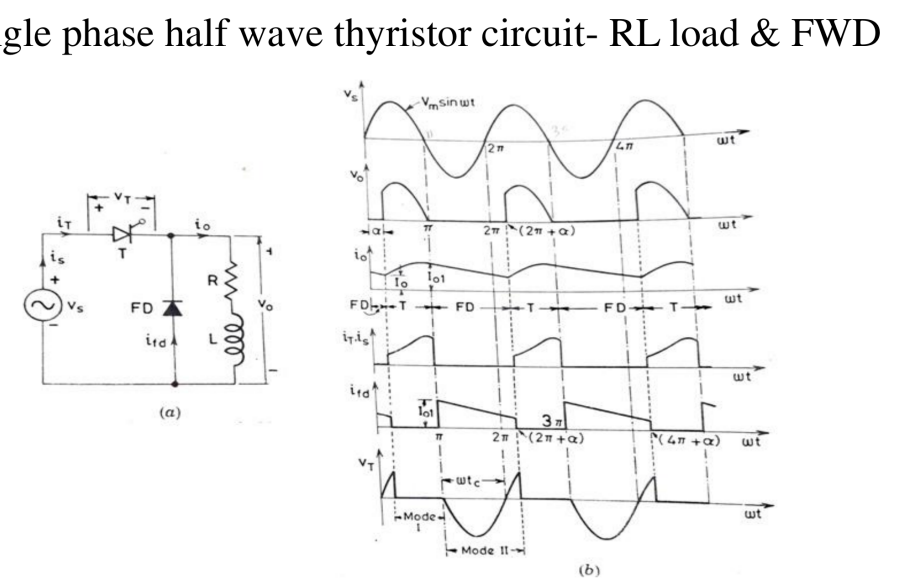

Figure 3: Single-phase half-wave circuit with RL load and Freewheeling Diode (FWD): Negative voltage is completely eliminated as current freewheels through D_{fw} from π to β.

Connecting an anti-parallel diode D_{fw} across the RL load completely eliminates the negative voltage excursion:

- At $\omega t = \pi$, as v_s goes negative, the cathode of D_{fw} becomes negative relative to its anode. D_{fw} turns ON automatically.
- D_{fw} clamps the load voltage to zero: $v_o(t) \approx 0$ for $\pi \leq \omega t \leq \beta$.
- The load current freewheels through R, L, and D_{fw}, decaying exponentially.
- The reverse voltage v_s immediately appears across the SCR, commutating it OFF at $\omega t = \pi$.

Active Load (RLE / Battery Charging)

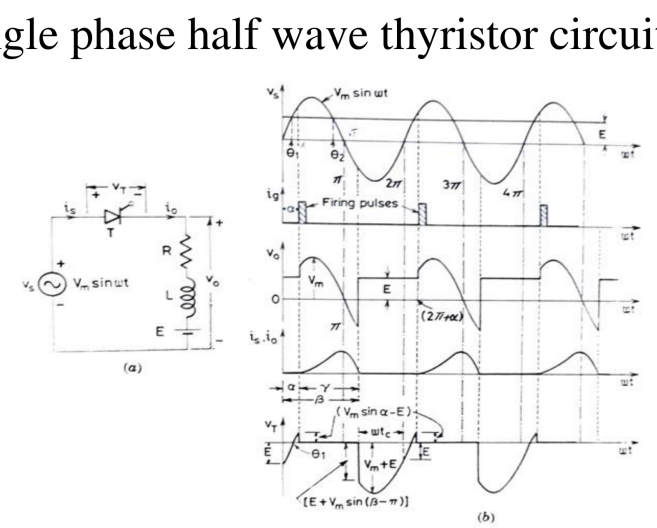

Figure 4: Single-phase half-wave circuit with RLE load: Thyristor is forward biased only when $v_s > E$. Firing angle α cannot be less than $\theta_1 = \sin^{-1}(E/V_m)$.

An RLE load models a practical DC motor drive or battery charger: R represents the winding/internal resistance, L is the motor armature inductance or smoothing inductor, and E is the motor back-EMF or battery terminal voltage.

Key difference from R and RL loads: The thyristor cannot conduct unless the supply voltage exceeds the back-EMF, i.e., $v_s > E$. This creates a *minimum firing angle* constraint:

$$\theta_1 = \sin^{-1}\left(\frac{E}{V_m}\right) \quad \Longrightarrow \quad \alpha \geq \theta_1$$

If we try to fire the SCR at $\alpha < \theta_1$, the net voltage $(v_s - E)$ across R and L is negative — no current can flow, and the gate pulse is wasted.

Once fired at $\alpha \geq \theta_1$, the load current builds up according to: $v_s - E = R\,i + L\,di/dt$. Current flows until the stored magnetic energy in L is fully dissipated (at extinction angle β). The current waveform shape depends on the L/R time constant and the magnitude of E.

Single-Phase Fully Controlled Bridge Rectifier (Full Converter)

A full converter uses **four thyristors** arranged in a bridge: T_1, T_2 in one diagonal and T_3, T_4 in the other. Unlike the half-wave circuit which uses only one half of each AC cycle, the full bridge rectifies *both* halves, doubling the average DC output.

Pure Resistive Load (R)

How the bridge works with R load:

- **Positive half cycle ($\alpha \leq \omega t \leq \pi$):** Thyristors T_1 and T_2 are fired simultaneously. Current flows: source $\rightarrow T_1 \rightarrow R \rightarrow T_2 \rightarrow$ source. Load voltage $v_o = v_s > 0$.
- At $\omega t = \pi$, supply reverses. With a resistive load, current drops to zero and T_1, T_2 turn OFF naturally.
- **Negative half cycle ($\pi + \alpha \leq \omega t \leq 2\pi$):** Thyristors T_3 and T_4 are fired. Current flows in the opposite path through the source but in the *same direction* through the load. Load voltage $v_o = -v_s > 0$.
- At 2π, supply reverses again, current drops to zero, and the cycle repeats.

The result is two positive voltage pulses per cycle, each of width $(\pi - \alpha)$. This is why the full converter produces twice the average voltage of a half-wave rectifier.

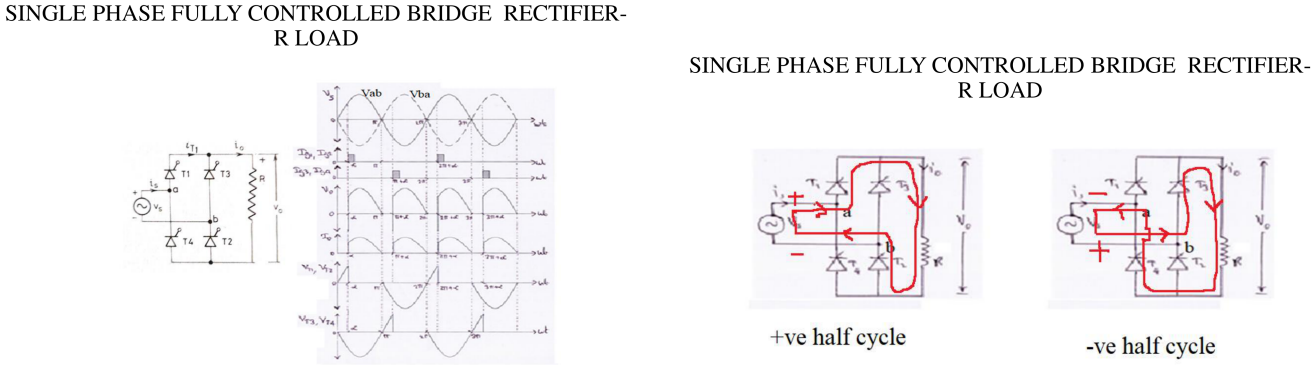

Figure 5: Single-phase fully controlled bridge with resistive load: Circuit schematic and synchronized voltage and current waveforms.

Continuous Ripple-Free Conduction (RL / RLE Load)

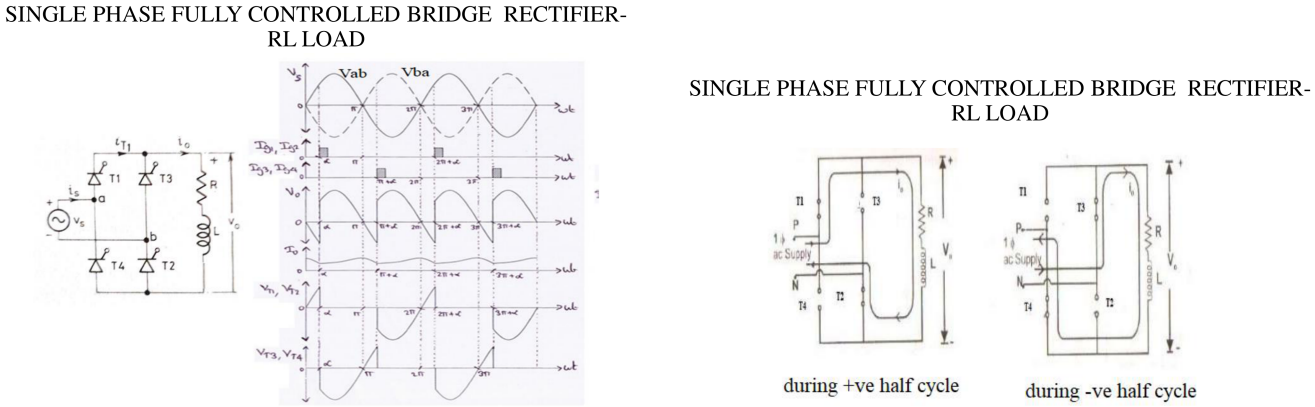

Figure 6: Single-phase fully controlled bridge with highly inductive load: Circuit schematic and continuous ripple-free waveforms showing negative voltage loops.

$$V_o = \frac{1}{\pi}\left[\int_{\alpha}^{\beta} V_m \sin \omega t \, d(\omega t) + E(\pi + \alpha - \beta)\right]$$

$$= \frac{V_m}{\pi}(\cos \alpha - \cos \beta) + E\left(1 - \frac{\gamma}{\pi}\right)$$

$$\gamma = \beta - \alpha = \text{conduction angle}$$

$$\text{Average load current, } I_o = \frac{V_o}{R} = \frac{V_m}{\pi R}(\cos \alpha - \cos \beta) + \frac{E}{R}\left(1 - \frac{\gamma}{\pi}\right)$$

Figure 7: Single-phase Full Converter detailed waveforms: Input voltage v_s, gate pulses $i_{G1,2}$ and $i_{G3,4}$, output voltage v_o, constant load current I_o, input line current i_s, thyristor currents, and device voltage v_{T1}.

Assuming load inductance is very large ($\omega L \gg R$), the load current is a smooth, continuous DC value I_o. This is the most important case for analysis.

The key idea: Unlike a resistive load (where current drops to zero when voltage reverses), a large inductor *forces the current to continue flowing* even when the supply voltage goes negative. The inductor "fights" any change in current — it simply will not allow I_o to drop. This creates negative voltage intervals across the load, which is the defining feature of a full converter with inductive load.

- **Interval 1 ($\alpha \leq \omega t \leq \pi$) — Normal Rectification:** Thyristors T_1, T_2 are fired at $\omega t = \alpha$. The supply voltage is positive, so $v_o = v_s > 0$. Current I_o flows through T_1, T_2. The inductor stores energy. Source current $i_s = +I_o$.
- **Interval 2 ($\pi \leq \omega t \leq \pi + \alpha$) — Inductor Holds On:** At $\omega t = \pi$, the supply voltage reverses and becomes negative. *But* T_1, T_2 cannot turn off — the enormous inductor maintains its current and keeps T_1, T_2 forward-biased. The load voltage goes negative: $v_o = v_s < 0$. During this interval, energy stored in the inductor is being returned to the AC source. Think of it as the inductor "pushing current uphill" against the reversed supply.
- **Interval 3 ($\pi + \alpha \leq \omega t \leq 2\pi$) — Commutation:** At $\omega t = \pi + \alpha$, thyristors T_3, T_4 receive their gate pulses and turn ON. Now the line voltage instantly reverse-biases T_1, T_2 (because T_3, T_4 connect the source in opposite polarity across T_1, T_2), forcing them OFF. This is natural (line) commutation. Now T_3, T_4 carry the load current, and $v_o = -v_s > 0$. Source current reverses: $i_s = -I_o$.
- **Interval 4 ($2\pi \leq \omega t \leq 2\pi + \alpha$) — Mirror of Interval 2:** T_3, T_4 continue conducting through the next zero crossing. Load voltage is again negative as the inductor holds conduction.

Derivation of Average Output Voltage (V_o)

Why integrate from α to $\pi + \alpha$? Look at the waveforms above. The output voltage v_o repeats every π radians (not 2π), because the full bridge rectifies both halves of the AC cycle. One complete output period runs from $\omega t = \alpha$ (when a thyristor pair is fired) to $\omega t = \pi + \alpha$ (when the next pair takes over). So the average is:

$$V_o = \frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} V_m \sin\omega t \, d(\omega t) = \frac{V_m}{\pi}[-\cos\omega t]_{\alpha}^{\pi+\alpha}$$

$$V_o = \frac{V_m}{\pi}[-\cos(\pi+\alpha) - (-\cos\alpha)] = \frac{V_m}{\pi}[\cos\alpha + \cos\alpha]$$

Single-Phase Full Converter Output Voltage

$$V_o = \frac{2V_m}{\pi}\cos\alpha = V_{o0}\cos\alpha$$

where $V_{o0} = \frac{2V_m}{\pi} \approx 0.6366 V_m$ is the maximum average output voltage at $\alpha = 0°$.
Physical insight: V_o is positive for $\alpha < 90°$ (rectification), zero at $\alpha = 90°$, and negative for $\alpha > 90°$ (inversion). The $\cos\alpha$ dependence means voltage control is non-linear — most sensitive near $\alpha = 90°$.

Derivation of RMS Output Voltage ($V_{o,\text{rms}}$)

$$V_{o,\text{rms}} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} V_m^2 \sin^2\omega t \, d(\omega t)} = \frac{V_m}{\sqrt{2}} = V_s$$

RMS Voltage Independence

For continuous conduction in a full converter, $V_{o,\text{rms}}$ **is always equal to the supply RMS voltage** V_s, regardless of firing angle α! This is because the integration of $\sin^2$ over any complete half-cycle always gives the same result. This means the output AC ripple power increases as α increases (since average DC power decreases but RMS stays constant).

Inverter Mode of Operation ($\alpha > 90°$)

LINE COMMUTATED INVERTER

- To achieve this in practice, a source of DC voltage which is equal to the average DC voltage must be connected at the output.
- This external source voltage (E) drives the direct current into the converter against the counter voltage produced at the output terminals.

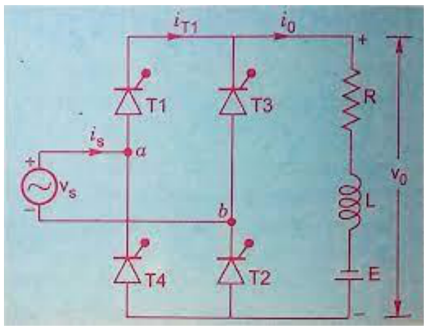

Figure 8: Line-Commutated Inverter Operation: When $\alpha > 90°$, average output voltage V_o becomes negative. An active DC source E connected in reverse polarity drives current through the thyristors, regenerating power into the AC line.

When would you want negative output voltage? Consider a DC motor decelerating: its back-EMF E acts as a source. By increasing α beyond $90°$, the converter output voltage reverses, allowing the motor's kinetic energy to flow

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-
RLE LOAD

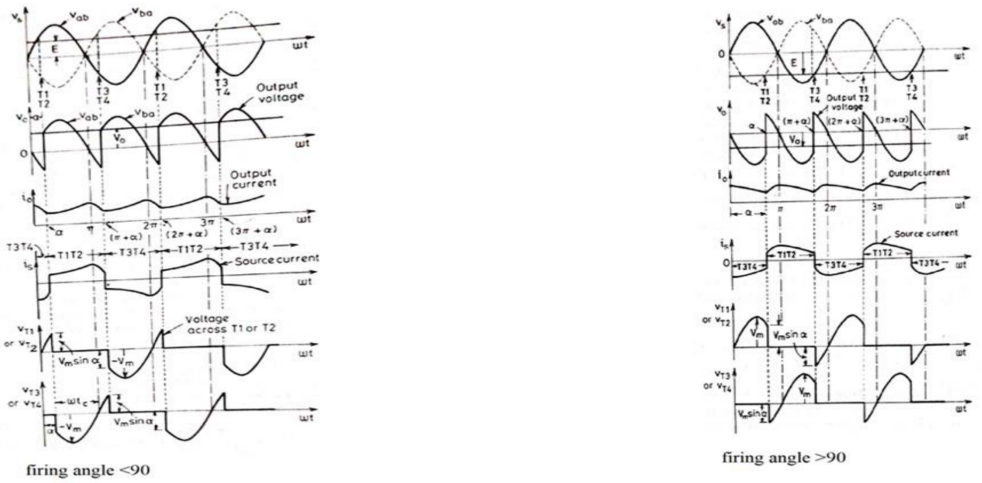

Figure 9: Single-phase full converter with RLE load: Detailed voltage and current waveforms in rectifying mode ($\alpha < 90°$) and inverting mode ($\alpha > 90°$).

back into the AC grid. This is called **regenerative braking**. The current direction stays the same (SCRs are unidirectional), but the voltage reverses — so power $P = V_o \times I_o$ becomes negative, meaning the load is returning power.

Commutation Failure in Inverter Mode

During inversion ($\alpha > 90°$), the reverse voltage available across the commutating thyristor exists only for the margin angle:

$$\gamma = 180° - \alpha$$

The thyristor requires a physical turn-off time t_q. If $\gamma < \omega t_q$, the outgoing thyristor fails to recover its forward blocking capability before v_s becomes positive. Both bridge legs conduct simultaneously, causing a massive short circuit known as **Commutation Failure**. To provide safety against this, practical firing angles are restricted to:

$$\alpha_{\text{max}} \approx 165° \text{ to } 170°$$

AC Source Performance Metrics (Fourier Analysis of Input Current)

Why is the input current a square wave? Recall from the interval analysis above: during $\alpha \leq \omega t \leq \pi + \alpha$, thyristors T_1, T_2 conduct and the source current is $i_s = +I_o$ (constant, because load current is ripple-free). Then during $\pi + \alpha \leq \omega t \leq 2\pi + \alpha$, thyristors T_3, T_4 conduct and the source current flips to $i_s = -I_o$. So the input current alternates between $+I_o$ and $-I_o$ — a perfect **square wave** of amplitude I_o, delayed by angle α from the supply voltage zero crossing.

Fourier Series Expansion

Since the square wave has odd symmetry and half-wave symmetry, only odd harmonics exist:

$$i_s(t) = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4I_o}{n\pi} \sin(n(\omega t - \alpha))$$

- **Fundamental Peak Current:** $I_{m1} = \frac{4I_o}{\pi}$
- **Fundamental RMS Current (I_{s1}):**

$$I_{s1} = \frac{4I_o}{\sqrt{2}\pi} = \frac{2\sqrt{2}}{\pi} I_o \approx 0.9003 I_o$$

- **Total RMS Input Current (I_s):**

$$I_s = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} I_o^2 \, d(\omega t)} = I_o$$

Power Factor Decomposition

Active power transmitted to the DC load is:

$$P = V_s I_{s1} \cos\phi_1 = V_s \left(\frac{2\sqrt{2}}{\pi} I_o \right) \cos\alpha = V_o I_o$$

where $\phi_1 = \alpha$ is the fundamental displacement angle.

Harmonic Performance Indices

1. **Displacement Factor (DF):** $DF = \cos\phi_1 = \cos\alpha$
2. **Current Distortion Factor (g):** $g = \frac{I_{s1}}{I_s} = \frac{\frac{2\sqrt{2}}{\pi} I_o}{I_o} = \frac{2\sqrt{2}}{\pi} \approx 0.9003$
3. **Input Power Factor (PF):** $PF = \frac{P}{S} = g \times DF = \frac{2\sqrt{2}}{\pi} \cos\alpha \approx 0.9003 \cos\alpha$
4. **Total Harmonic Distortion (THD):** $THD = \sqrt{\frac{I_s^2 - I_{s1}^2}{I_{s1}^2}} = \sqrt{\frac{1}{g^2} - 1} = \sqrt{\frac{\pi^2}{8} - 1} \approx 48.34\%$

Single-Phase Half-Controlled Converter (Semi-Converter)

What is a Semi-Converter and why use it? A full converter uses four thyristors, requiring four separate gate trigger channels with pulse transformers or optocouplers. However, many practical industrial DC loads (like battery chargers, electroplating, and unidirectional DC motor drives) only operate in **Quadrant I** ($V_o > 0, I_o > 0$) and never need regenerative inversion.

In such applications, replacing two thyristors with passive uncontrolled power diodes creates a **Semi-Converter (Half-Controlled Bridge)**. This provides several major engineering advantages:

1. **Lower Cost & Simpler Controls:** Diodes are cheaper than SCRs and require no gate drive circuits, power supplies, or synchronisation logic.

2. **Built-in Freewheeling Action:** The diodes inherently provide a freewheeling path across the load whenever the AC voltage reverses, completely eliminating negative output voltage loops without needing a separate external freewheeling diode.

3. **Superior Power Factor & Reduced Reactive Power:** Because load current freewheels during $\pi \leq \omega t \leq \pi + \alpha$, the AC supply draws zero current during this interval ($i_s = 0$). This reduces the reactive power drawn from the grid and gives a significantly higher power factor than a full converter at all firing angles $\alpha > 0$.

Half controlled converter with RL load (Symmetrical)

Asymmetrical semi converter with RL load

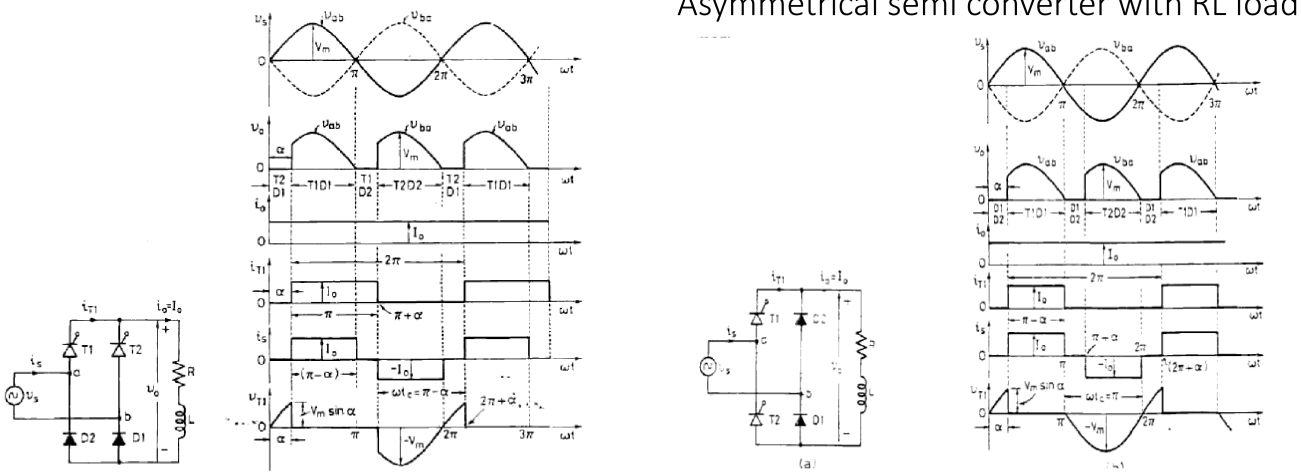

Figure 10: Single-phase Semi-Converter configurations: (Left) Symmetrical bridge with thyristors in top bus and diodes in bottom bus. (Right) Asymmetrical bridge with one thyristor and one diode per leg.

Symmetrical vs. Asymmetrical Configurations

- **Symmetrical Bridge:** Thyristors T_1, T_2 are placed in the positive (top) arm, and diodes D_1, D_2 in the negative (bottom) arm. During freewheeling, current circulates through one thyristor and one diode (e.g., T_1 and D_2). Therefore, the thyristor must carry current during the freewheeling period, which prolongs its conduction angle to $(\pi + \alpha)$ and gives a circuit turn-off time of $t_c = (\pi - \alpha)/\omega$.

- **Asymmetrical Bridge:** Each of the two bridge legs contains one thyristor and one diode (T_1, D_1 in leg 1, and T_2, D_2 in leg 2). Here, freewheeling occurs strictly through the two series diodes (D_1 and D_2). Both thyristors turn off cleanly at $\omega t = \pi$, providing a full half-cycle turn-off margin $t_c = \pi/\omega$, which makes commutation much more robust.

Operating Intervals & Mathematical Derivation of V_o

Assuming continuous, ripple-free load current I_o:

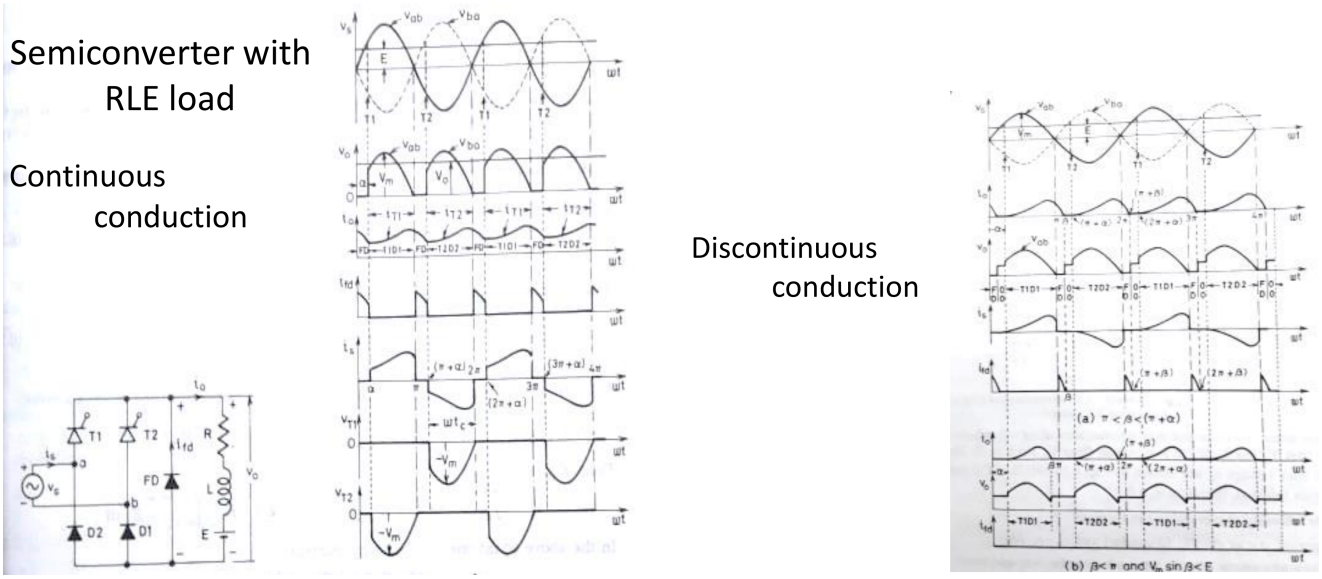

Figure 11: Semi-Converter Waveforms: (Left) Continuous conduction mode showing voltage clamped to zero during freewheeling intervals (π to $\pi + \alpha$). (Right) Discontinuous conduction mode.

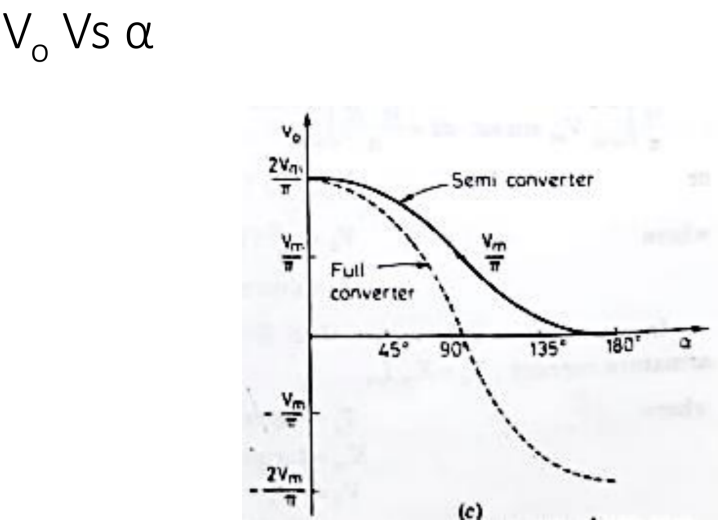

Figure 12: Normalized Average Output Voltage (V_o/V_{o0}) vs Firing Angle (α): Comparison showing that the full converter can produce negative voltage (inverting mode), while the semi-converter is strictly non-negative ($V_o \geq 0$).

- **Interval 1 ($\alpha \leq \omega t \leq \pi$) — Power Delivery:** Thyristor T_1 and diode D_1 conduct. The load is connected directly to the AC source: $v_o(t) = v_s(t) = V_m \sin \omega t$. Active power flows from the AC utility grid into the DC load. Source current is $i_s = +I_o$.
- **Interval 2 ($\pi \leq \omega t \leq \pi + \alpha$) — Freewheeling Mode:** At $\omega t = \pi$, supply voltage v_s crosses zero and turns negative. Diode D_2 immediately becomes forward-biased by the collapsing inductor EMF. Current shifts from the AC source into the freewheeling path (D_1, D_2). The load terminals are short-circuited through the conducting diodes, clamping the load voltage to zero: $v_o(t) = 0$. During this interval, the AC source is disconnected: $i_s(t) = 0$.

Average Output Voltage Derivation: Since the output voltage is $V_m \sin \omega t$ from α to π and 0 from π to $\pi + \alpha$, we integrate over the repeating half-cycle period π:

$$V_o = \frac{1}{\pi}\left[\int_{\alpha}^{\pi} V_m \sin \omega t \, d(\omega t) + \int_{\pi}^{\pi+\alpha} 0 \, d(\omega t)\right] = \frac{V_m}{\pi}[-\cos \omega t]_{\alpha}^{\pi}$$

$$V_o = \frac{V_m}{\pi}[-\cos \pi - (-\cos \alpha)] = \frac{V_m}{\pi}[1 + \cos \alpha]$$

Single-Phase Semi-Converter Output Voltage

$$V_o = \frac{V_m}{\pi}(1+\cos\alpha) = \frac{\sqrt{2}V_s}{\pi}(1+\cos\alpha)$$

- At $\alpha = 0°$: $V_o = \frac{2V_m}{\pi}$ (same as full converter at $\alpha = 0$).
- At $\alpha = 90°$: $V_o = \frac{V_m}{\pi}$ (half the maximum DC voltage).
- At $\alpha = 180°$: $V_o = 0$. Because $(1+\cos\alpha) \geq 0$ for all $\alpha \in [0, \pi]$, V_o can **never be negative**!

AC Source Current & Power Factor Advantage

The source current $i_s(t)$ flows only during the active conduction intervals (α to π, and $\pi+\alpha$ to 2π), each of angular width $(\pi - \alpha)$:

- **RMS Supply Current:**

$$I_s = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi} I_o^2 \, d(\omega t)} = I_o\sqrt{\frac{\pi-\alpha}{\pi}}$$

  Notice that as α increases, I_s drops because the current freewheels for longer periods.

- **Displacement Factor (DF):** The fundamental input current is centered in the conduction interval $[\alpha, \pi]$, meaning its phase shift relative to supply voltage is only $\phi_1 = \alpha/2$:

$$DF = \cos\left(\frac{\alpha}{2}\right)$$

  Compare this to a full converter where $DF = \cos\alpha$. For example, at $\alpha = 60°$, full converter $DF = 0.50$, whereas semi-converter $DF = \cos 30° = 0.866$ — a 73% improvement!

- **Input Power Factor (PF):**

$$PF = \frac{P}{S} = \frac{V_o I_o}{V_s I_s} = \frac{\frac{V_m}{\pi}(1+\cos\alpha)I_o}{V_s\left(I_o\sqrt{\frac{\pi-\alpha}{\pi}}\right)} = \frac{\sqrt{2}(1+\cos\alpha)}{\sqrt{\pi(\pi-\alpha)}}$$

Effect of Source Inductance (L_s) on Controlled Rectifiers

In practical installations, the AC source possesses leakage inductance from feeding transformers and supply distribution lines (L_s).

Physical Mechanism of Commutation Overlap

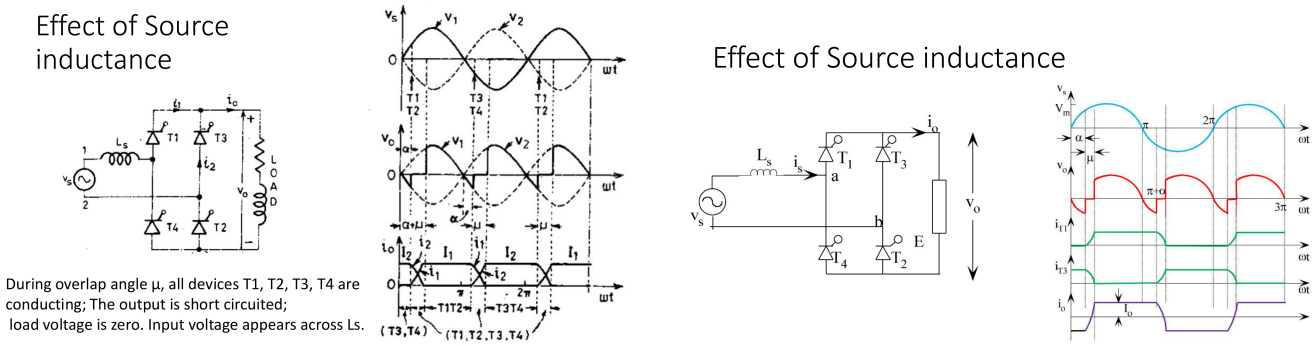

Figure 13: Effect of Source Inductance (L_s): (Left) Circuit schematic showing supply inductance L_s. (Right) Commutation overlap angle μ during which all 4 thyristors conduct simultaneously and output voltage is shorted ($v_o = 0$).

Because of L_s, the line current cannot change instantaneously:

$$v_{L_s} = L_s \frac{di_s}{dt}$$

When incoming thyristors T_1, T_2 are triggered at $\omega t = \alpha$, outgoing thyristors T_3, T_4 cannot shut off abruptly. For an angular duration μ (the **overlap angle**), **all four thyristors conduct simultaneously**. During this interval, the AC source is shorted through L_s and the bridge switches, causing the output voltage to be **zero**:

$$v_o(t) = 0 \quad \text{for } \alpha \leq \omega t \leq \alpha + \mu$$

Mathematical Derivation of Overlap Angle μ

During overlap ($\alpha \leq \omega t \leq \alpha + \mu$): Let i_1 be incoming SCR current and i_2 be outgoing SCR current:

$$i_1 + i_2 = I_o \implies \frac{di_2}{dt} = -\frac{di_1}{dt}$$

Line current is $i_s = i_1 - i_2$. Therefore:

$$\frac{di_s}{dt} = \frac{di_1}{dt} - \frac{di_2}{dt} = 2\frac{di_1}{dt}$$

KVL around the AC source loop gives:

$$v_s(t) = L_s \frac{di_s}{dt} = 2L_s \frac{di_1}{dt} \implies \frac{di_1}{d(\omega t)} = \frac{V_m}{2\omega L_s} \sin \omega t$$

Integrating from $\omega t = \alpha$ ($i_1 = 0$) to $\omega t = \alpha + \mu$ ($i_1 = I_o$):

$$\int_0^{I_o} di_1 = \frac{V_m}{2\omega L_s} \int_\alpha^{\alpha+\mu} \sin \omega t \, d(\omega t)$$

$$I_o = \frac{V_m}{2\omega L_s} [\cos \alpha - \cos(\alpha + \mu)]$$

Commutation Overlap Equation

$$\cos\alpha - \cos(\alpha+\mu) = \frac{2\omega L_s I_o}{V_m} = \frac{\sqrt{2}\omega L_s I_o}{V_s}$$

Average Output Voltage Reduction & Equivalent Commutation Resistance

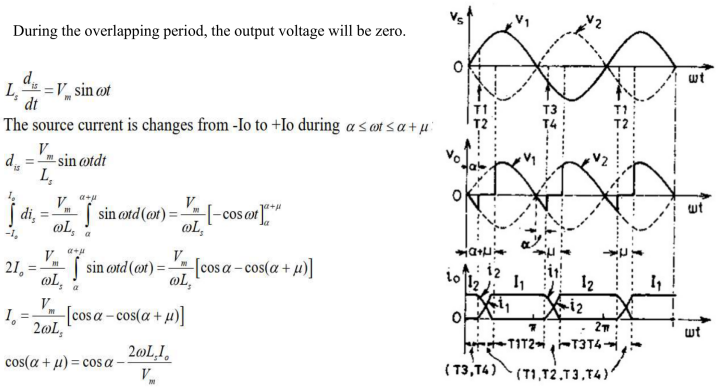

Figure 14: Detailed voltage and current commutation waveforms: Commutation interval μ, decay of outgoing current i_2, rise of incoming current i_1, and the resulting lost volt-second area ΔV_o.

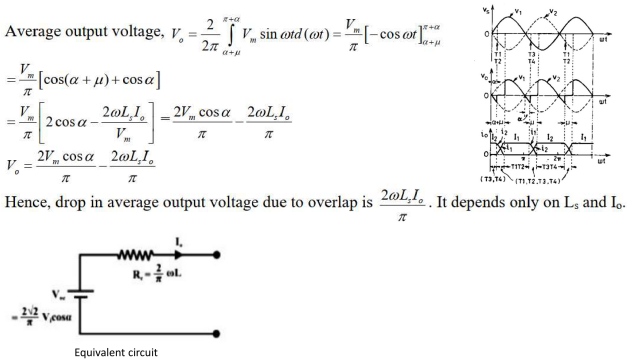

Figure 15: DC Equivalent Circuit with Source Inductance: Converter modeled as ideal DC voltage source $V_{o0} \cos \alpha$ in series with equivalent commutation resistance $R_c = 2fL_s$.

Since $v_o(t) = 0$ during overlap, conduction only provides voltage from $\alpha + \mu$ to $\pi + \alpha$:

$$V_o = \frac{1}{\pi} \int_{\alpha+\mu}^{\pi+\alpha} V_m \sin \omega t \, d(\omega t) = \frac{V_m}{\pi} [\cos \alpha + \cos(\alpha + \mu)]$$

Substituting $\cos(\alpha + \mu) = \cos \alpha - \frac{2\omega L_s I_o}{V_m}$:

$$V_o = \frac{V_m}{\pi} \left[2 \cos \alpha - \frac{2\omega L_s I_o}{V_m} \right] = \frac{2V_m}{\pi} \cos \alpha - \frac{\omega L_s}{\pi} I_o$$

Output Voltage with Source Inductance

$$V_o = \frac{2V_m}{\pi} \cos \alpha - \Delta V_o = V_{o0} \cos \alpha - R_c I_o$$

where:

$$\Delta V_o = \frac{\omega L_s}{\pi} I_o = 2fL_s I_o, \qquad R_c = \frac{\omega L_s}{\pi} = 2fL_s \quad (\Omega)$$

Nature of Equivalent Commutation Resistance R_c

R_c is an **apparent** or **fictitious resistance**. It represents the loss of average voltage area due to overlap. **No thermal heat loss ($I_o^2 R_c$) occurs in R_c!**

Three-Phase Controlled Rectifiers

Three-phase rectifiers provide higher DC power capability, lower ripple factor, and a ripple frequency of $6f_s = 300$ Hz (for 50 Hz supply).

Three-Phase Voltage Relationships — Quick Reference

For a balanced three-phase supply with RMS line-to-line voltage V_L and RMS phase voltage V_{ph}:

$$\underbrace{V_{mp}}_{\text{Peak Phase}} = \sqrt{2}\,V_{ph}, \qquad V_L = \sqrt{3}\,V_{ph}, \qquad \underbrace{V_{ml}}_{\text{Peak Line}} = \sqrt{2}\,V_L = \sqrt{3}\,V_{mp}$$

Example: For $V_L = 400$ V: $V_{ph} = 231$ V, $V_{mp} = 326.6$ V, $V_{ml} = 565.7$ V.

Three-Phase Half-Wave Converter (3-Pulse Rectifier)

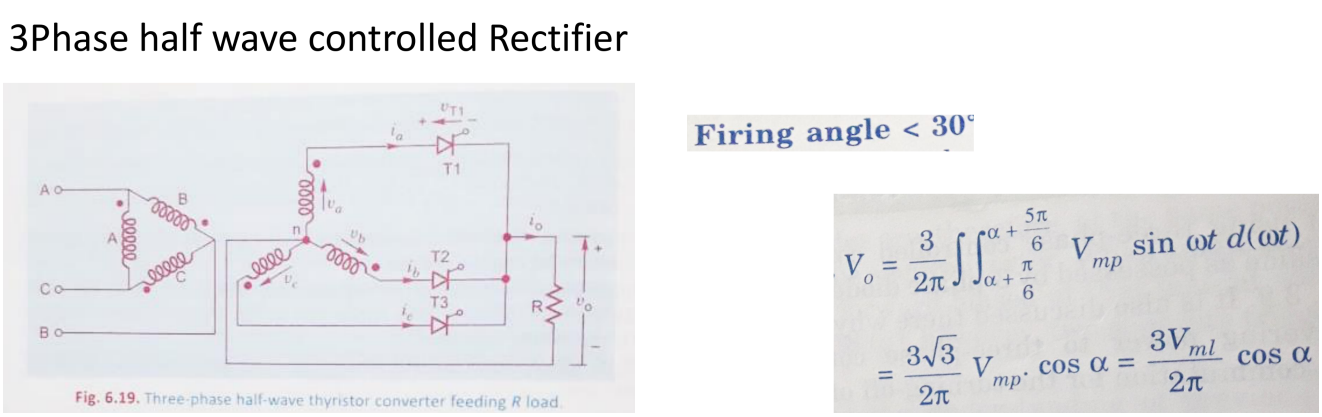

Figure 16: Three-phase half-wave controlled rectifier: (Left) Circuit schematic with 3 thyristors. (Right) Output voltage waveform for $\alpha = 0°$.

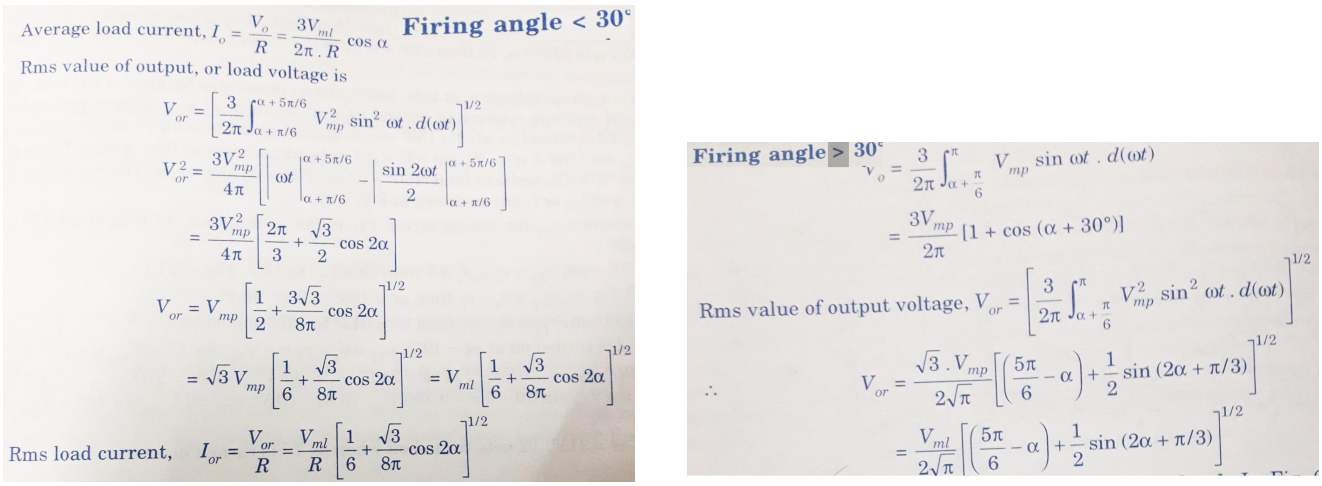

Figure 17: Three-phase half-wave waveforms at delayed firing: (Left) $\alpha = 30°$. (Right) $\alpha = 60°$.

Each thyristor conducts for 120° ($2\pi/3$ radians). There are 3 pulses per cycle of period $2\pi/3$. Integrating the phase voltage over one conduction interval:

$$V_o = \frac{1}{2\pi/3}\int_{-\pi/6+\alpha}^{\pi/2+\alpha} V_{mp}\sin\omega t\, d(\omega t) = \frac{3V_{mp}}{2\pi}\left[-\cos\omega t\right]_{-\pi/6+\alpha}^{\pi/2+\alpha}$$

Using $\cos(-\pi/6+\alpha)-\cos(\pi/2+\alpha)=2\cos\alpha\sin(60^\circ)=\sqrt{3}\cos\alpha$:

3-Phase Half-Wave (3-Pulse) Output Voltage

$$V_o=\frac{3\sqrt{3}\,V_{mp}}{2\pi}\cos\alpha=\frac{3\,V_{ml}}{2\pi}\cos\alpha\approx 1.17\,V_{ph}\cos\alpha$$

where V_{mp} is the peak phase voltage and $V_{ml}=\sqrt{3}\,V_{mp}$ is the peak line-to-line voltage.

Three-Phase Fully Controlled 6-Pulse Bridge Converter

Contains 6 thyristors:

- **Positive Group (Cathodes common to DC+ bus):** T_1 (phase a), T_3 (phase b), T_5 (phase c).
- **Negative Group (Anodes common to DC− bus):** T_4 (phase a), T_6 (phase b), T_2 (phase c).

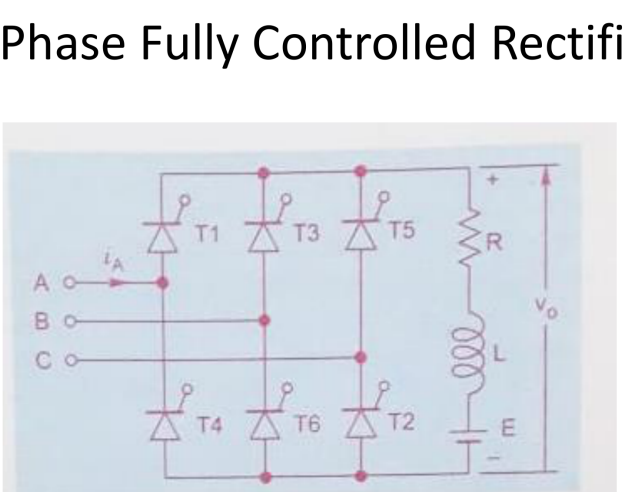

Figure 18: Three-Phase Fully Controlled Bridge Converter (6-Pulse): Schematic diagram showing positive group (T_1, T_3, T_5) and negative group (T_4, T_6, T_2).

Firing Sequence and Conduction Table

Thyristors are triggered in numerical order at 60° intervals:

$$T_1 \longrightarrow T_2 \longrightarrow T_3 \longrightarrow T_4 \longrightarrow T_5 \longrightarrow T_6 \longrightarrow T_1$$

Each SCR conducts for 120°. At any instant, two devices conduct (one positive group, one negative group).

Table 1: Conduction Intervals and Line Voltages for 3-Phase Full Converter

Interval (ωt)	Conducting Pair	Output Voltage $v_o(t)$
$60° + \alpha$ to $120° + \alpha$	T_1, T_6	$v_{ab} = v_{an} - v_{bn}$
$120° + \alpha$ to $180° + \alpha$	T_1, T_2	$v_{ac} = v_{an} - v_{cn}$
$180° + \alpha$ to $240° + \alpha$	T_3, T_2	$v_{bc} = v_{bn} - v_{cn}$
$240° + \alpha$ to $300° + \alpha$	T_3, T_4	$v_{ba} = v_{bn} - v_{an}$
$300° + \alpha$ to $360° + \alpha$	T_5, T_4	$v_{ca} = v_{cn} - v_{an}$
$360° + \alpha$ to $60° + \alpha$	T_5, T_6	$v_{cb} = v_{cn} - v_{bn}$

Waveform Evolution with Firing Angle α

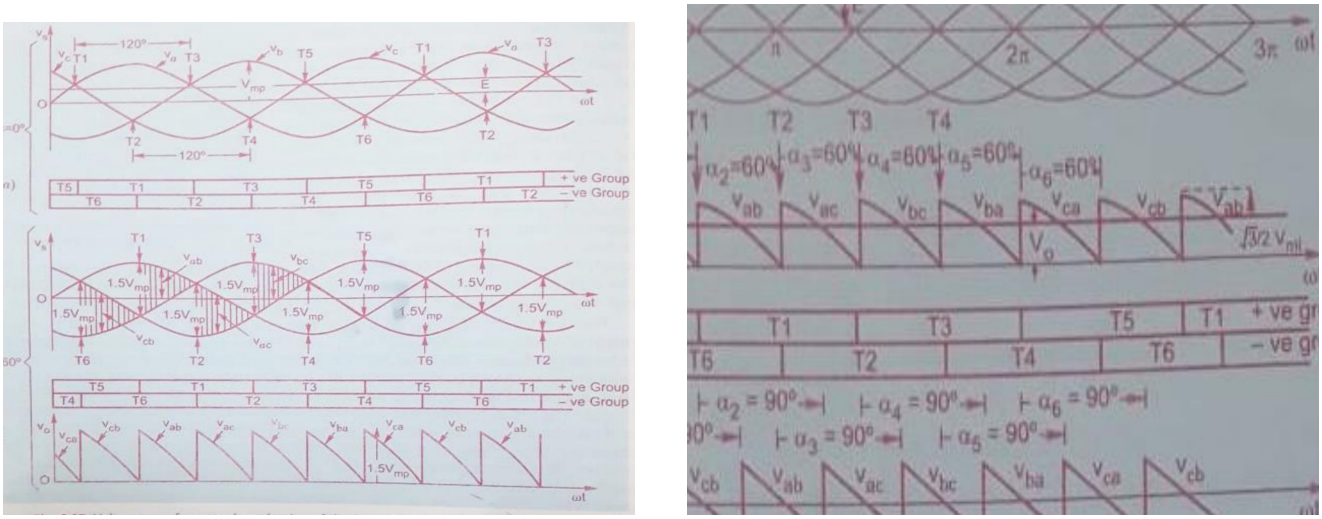

Figure 19: Three-phase Full Converter Output Voltage Waveforms: (Left) $\alpha = 0°$ (pure 6-pulse diode envelope). (Right) $\alpha = 30°$.

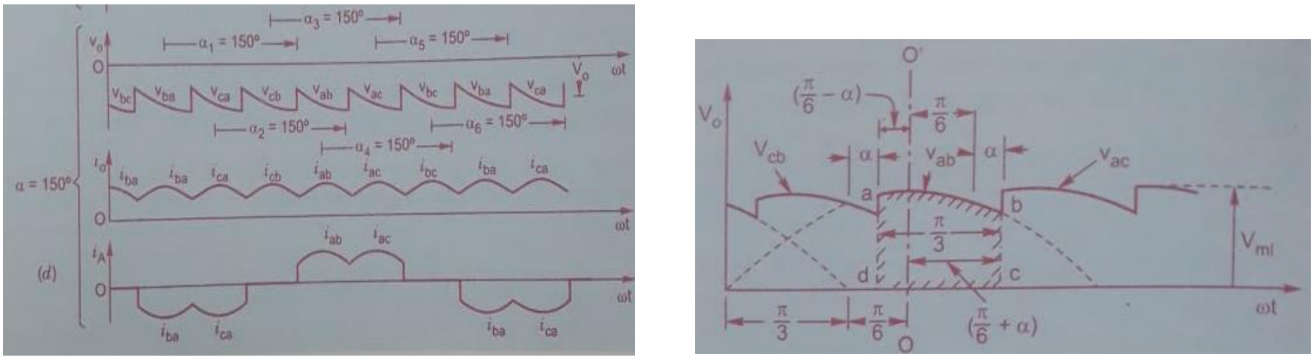

Figure 20: Three-phase Full Converter Waveforms: (Left) $\alpha = 60°$ (instantaneous voltage touches zero). (Right) $\alpha = 90°$ (voltage swings equally positive and negative, average $V_o = 0$).

Derivation of Average Output Voltage (V_o)

There are 6 symmetric pulses per cycle of duration $\pi/3$ (60°):

$$V_o = \frac{1}{\pi/3}\int_{-\pi/6+\alpha}^{\pi/6+\alpha} V_{ml}\cos\omega t\, d(\omega t) = \frac{3V_{ml}}{\pi}[\sin\omega t]_{-\pi/6+\alpha}^{\pi/6+\alpha}$$

Using $\sin(A+B) - \sin(A-B) = 2\cos A \sin B$:

$$V_o = \frac{3V_{ml}}{\pi}\left[2\cos\alpha\sin(30°)\right]$$

3-Phase Full Converter Output Voltage

$$V_o = \frac{3V_{ml}}{\pi}\cos\alpha = \frac{3\sqrt{3}V_{mp}}{\pi}\cos\alpha = \frac{3\sqrt{2}V_L}{\pi}\cos\alpha \approx 1.35V_L\cos\alpha$$

Three-Phase Half-Controlled Converter (Semi-Converter)

Circuit Topology & Operating Philosophy: The 3-phase semi-converter replaces the 3 thyristors in the negative group with 3 power diodes (D_4, D_6, D_2), while retaining 3 thyristors (T_1, T_3, T_5) in the positive group (or vice versa).

- **Why use it?** It reduces the number of gate firing circuits from 6 down to 3, significantly lowering control complexity and hardware cost.
- **Single-Quadrant Operation:** Because of the passive diodes, the DC terminal voltage V_o can **never reverse polarity** ($V_o \geq 0$). Thus, it operates strictly in **Quadrant I** as a controlled rectifier.
- **Higher Power Factor:** Just like the 1-phase semi-converter, the diodes provide an automatic freewheeling action that clamps negative voltage excursions, eliminating reactive power circulation and yielding a higher power factor than the 6-pulse full bridge for $\alpha > 0$.

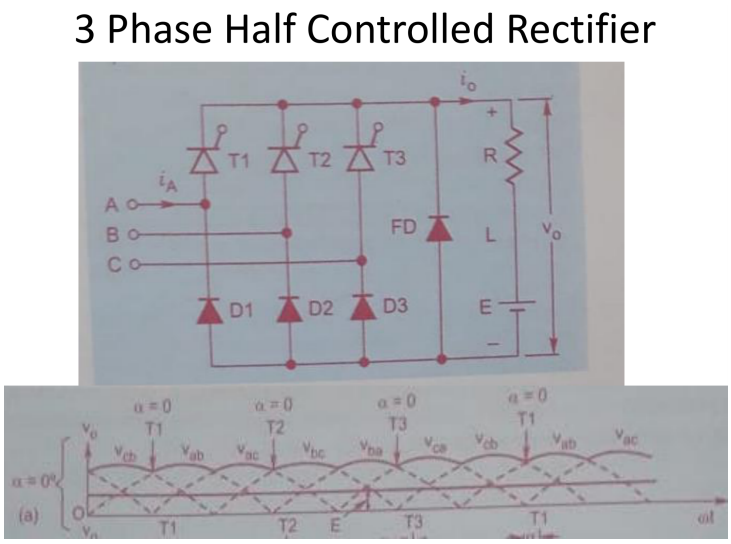

Figure 21: Three-Phase Semi-Converter: Bridge topology with 3 thyristors in positive group and 3 diodes in negative group.

The Critical Transition at $\alpha = 60°$

The operation of the 3-phase semi-converter bifurcates into two completely distinct operational regimes:

1. **Continuous 6-Pulse Conduction ($\alpha \leq 60°$):** The output voltage never touches zero. At any instant, one thyristor from the top group and one diode from the bottom group (belonging to different phases) conduct. The output voltage is composed of 6 distinct line voltage segments per cycle, with a ripple frequency of $6f_s = 300$ Hz. Freewheeling does not occur because the instantaneous output line voltage remains strictly positive.
2. **Freewheeling 3-Pulse Conduction ($\alpha > 60°$):** As α exceeds 60°, the instantaneous line voltage tries to swing negative. The moment it reaches zero, the diode in the same phase leg as the conducting thyristor turns ON. Current now circulates (freewheels) internally through that thyristor and diode pair in the same leg!

   During this freewheeling interval, the load terminals are short-circuited through the leg, clamping the output voltage to **zero** ($v_o = 0$) and disconnecting the AC line current ($i_s = 0$). Three line voltage pulses disappear from the output envelope, reducing the ripple frequency to $3f_s = 150$ Hz.

Derivation of Average Output Voltage (V_o)

Each of the 3 thyristors conducts for 120° ($2\pi/3$). Taking the line voltage as reference ($v_{line} = V_{ml} \sin \omega t$), active conduction delivers voltage from $\omega t = \alpha$ up to π:

$$V_o = \frac{3}{2\pi} \int_{\alpha}^{\pi} V_{ml} \sin \omega t \, d(\omega t) = \frac{3V_{ml}}{2\pi} [-\cos \omega t]_{\alpha}^{\pi} = \frac{3V_{ml}}{2\pi} [-\cos \pi - (-\cos \alpha)]$$

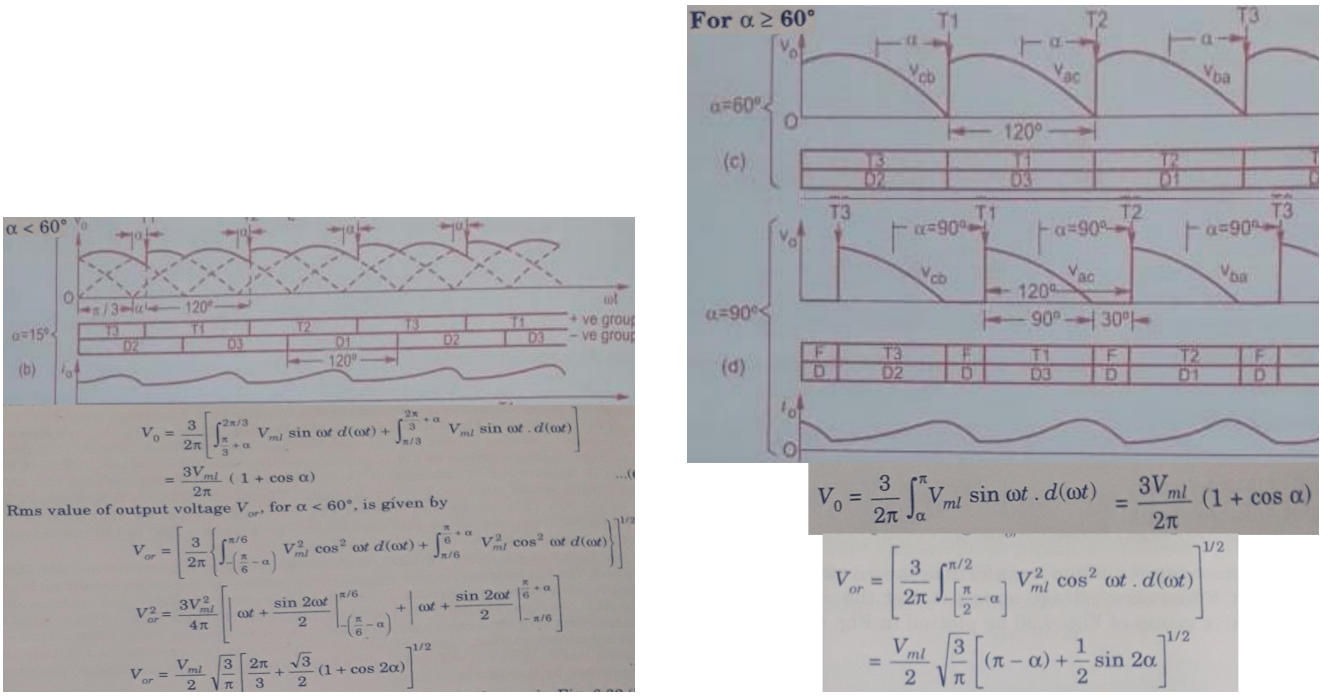

Figure 22: Three-Phase Semi-Converter Waveforms: (Left) $\alpha = 0°$ (6-pulse operation). (Right) $\alpha = 60°$ (transition to 3-pulse freewheeling operation).

3-Phase Semi-Converter Output Voltage

$$V_o = \frac{3V_{ml}}{2\pi}(1+\cos\alpha) = \frac{3\sqrt{3}V_{mp}}{2\pi}(1+\cos\alpha) = \frac{3\sqrt{2}V_L}{2\pi}(1+\cos\alpha) \approx 0.675V_L(1+\cos\alpha)$$

- At $\alpha = 0°$: $V_o = \frac{3V_{ml}}{\pi} \approx 1.35V_L$ (identical to 6-pulse full converter).
- At $\alpha = 60°$: $V_o = \frac{3V_{ml}}{2\pi}(1+0.5) = \frac{4.5V_{ml}}{2\pi} \approx 1.013V_L$.
- At $\alpha = 180°$: $V_o = 0$ V.

Source Inductance in Three-Phase Full Converter

Physical Commutation Process in 3-Phase Bridges: In a 3-phase full converter, commutation occurs **6 times per AC supply cycle** (once every 60°). When an incoming thyristor (say T_3 in phase b) is fired at α, the outgoing thyristor (T_1 in phase a) cannot shut off instantaneously because the supply inductance L_s in phase a opposes current collapse ($v_{Ls} = L_s di/dt$).

Commutation Overlap Mechanics

During the overlap interval μ:

- **Three devices conduct simultaneously:** The two commutating devices in one group (T_1 and T_3) plus the one return device in the opposite group (T_2 in phase c).
- **Phase Line Short-Circuit:** Phases a and b are tied together through their source inductances L_s. The potential of the positive DC rail is the midpoint voltage: $v_{pos} = \frac{v_a+v_b}{2}$.
- The output line voltage during overlap is reduced by $\frac{v_a-v_b}{2} = \frac{v_{ab}}{2}$.
- This loss of volt-second area causes a net reduction ΔV_o in the average DC output voltage.

Derivation of Overlap Angle μ and Voltage Drop ΔV_o

During overlap of phase a and phase b:

$$v_a - L_s\frac{di_a}{dt} = v_b - L_s\frac{di_b}{dt} \implies v_{ab} = L_s\left(\frac{di_a}{dt} - \frac{di_b}{dt}\right)$$

Since $i_a + i_b = I_o$ (constant DC current), $\frac{di_a}{dt} = -\frac{di_b}{dt}$. Thus:

$$v_{ab} = 2L_s\frac{di_b}{dt} = V_{ml}\sin\omega t$$

Integrating incoming current i_b from 0 to I_o as ωt goes from α to $\alpha+\mu$:

$$\int_0^{I_o} di_b = \frac{V_{ml}}{2\omega L_s}\int_\alpha^{\alpha+\mu}\sin\omega t\, d(\omega t) \implies I_o = \frac{V_{ml}}{2\omega L_s}[\cos\alpha - \cos(\alpha+\mu)]$$

3-Phase Overlap Angle & Equivalent Resistance

$$\cos\alpha - \cos(\alpha+\mu) = \frac{2\omega L_s I_o}{V_{ml}} = \frac{\sqrt{2}\omega L_s I_o}{V_L}$$

Because there are **6 commutations per cycle**, the average voltage drop across the period 2π is:

$$\Delta V_o = \frac{6\times(\text{lost volt-second area per commutation})}{2\pi} = \frac{6\times(\omega L_s I_o)}{2\pi} = \frac{3\omega L_s}{\pi}I_o = 3fL_sI_o$$

Notice that in 3-phase, $\Delta V_o = 3fL_sI_o$, whereas in 1-phase it is $2fL_sI_o$ (proportional to the number of commutations per cycle!).

$$R_c = \frac{3\omega L_s}{\pi} = 3fL_s \quad (\Omega)$$

$$V_o = \frac{3V_{ml}}{\pi}\cos\alpha - \Delta V_o = \frac{3\sqrt{2}V_L}{\pi}\cos\alpha - 3fL_sI_o = \frac{3V_{ml}}{2\pi}[\cos\alpha + \cos(\alpha+\mu)]$$

DC-DC Switching Regulators (Buck, Boost, Buck-Boost)

Switch-mode converters convert an unregulated DC input V_d into a regulated output V_o with theoretical efficiency of 100% by using switches in saturation or cutoff.

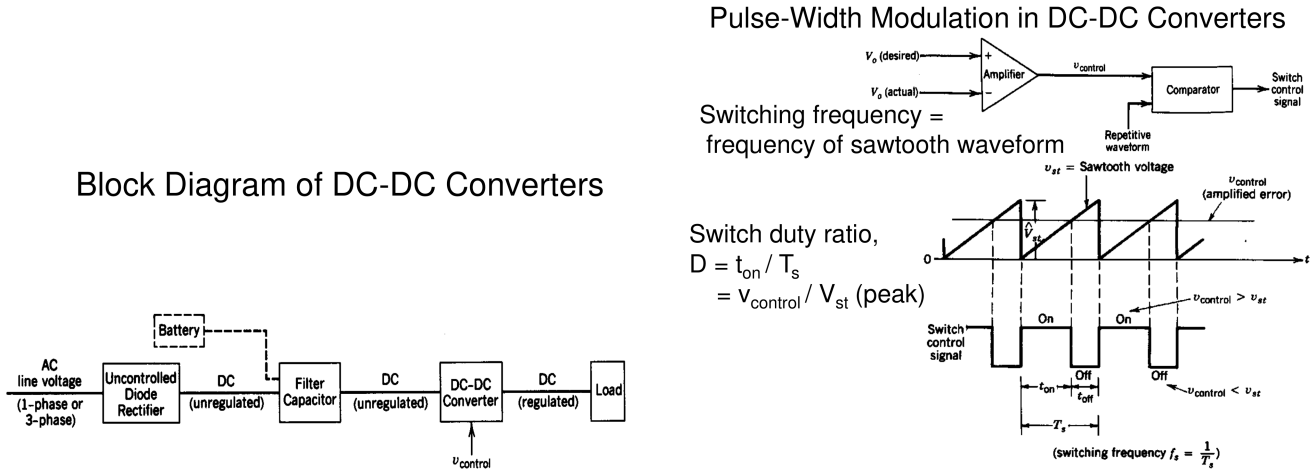

Figure 23: DC-DC Switch-Mode Conversion: (Left) Functional block diagram. (Right) Pulse-Width Modulation (PWM) duty cycle control by comparing control voltage with sawtooth carrier.

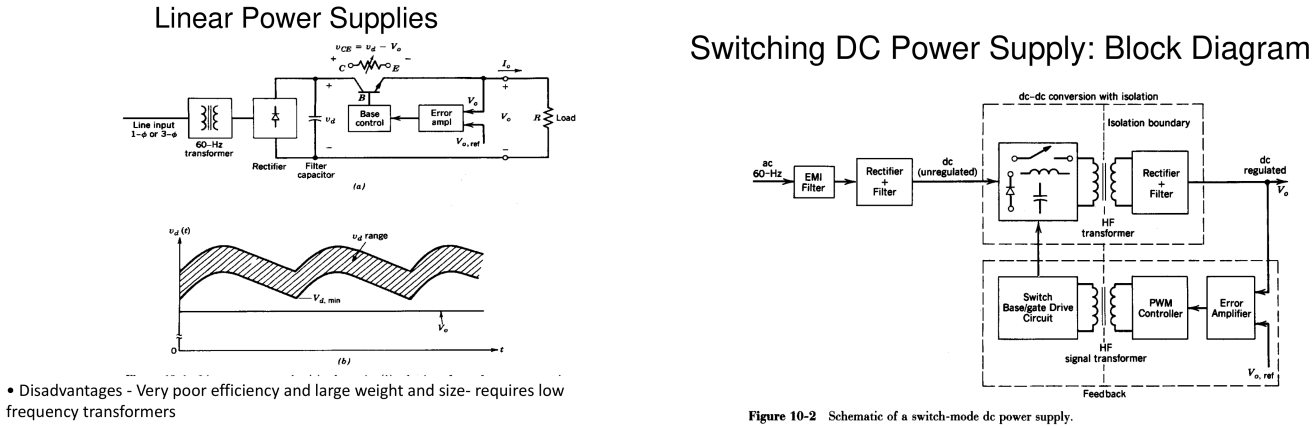

Figure 24: Power Supply Comparison: (Left) Linear regulator dissipates $(V_{\text{in}} - V_o)I_o$ as continuous heat. (Right) Switching power supply operates transistor as an efficient high-frequency switch.

Two Fundamental Steady-State Laws

Conservation Principles in Power Electronics

1. **Inductor Volt-Second Balance:** Because $v_L = L\frac{di_L}{dt}$ and $i_L(T_s) = i_L(0)$:

$$\int_0^{T_s} v_L(t)\,dt = 0 \iff \langle v_L \rangle = 0$$

2. **Capacitor Charge Balance:** Because $i_C = C\frac{dv_C}{dt}$ and $v_C(T_s) = v_C(0)$:

$$\int_0^{T_s} i_C(t)\,dt = 0 \iff \langle i_C \rangle = 0$$

The Small-Ripple Approximation (Key Analytical Assumption)

In steady-state CCM operation, the inductor current $i_L(t)$ and capacitor voltage $v_C(t)$ contain small triangular AC ripples superimposed on large DC values. The **small-ripple approximation** assumes:

$$i_L(t) \approx I_L \quad \text{(average value)}, \qquad v_C(t) \approx V_o \quad \text{(average value)}$$

This allows us to treat i_L and v_C as constants when writing KVL/KCL during the ON and OFF intervals, greatly simplifying the analysis. This approximation is valid when $\Delta I_L \ll I_L$ and $\Delta V_o \ll V_o$, which holds whenever the filter L and C are properly designed.

Step-Down (Buck) Converter

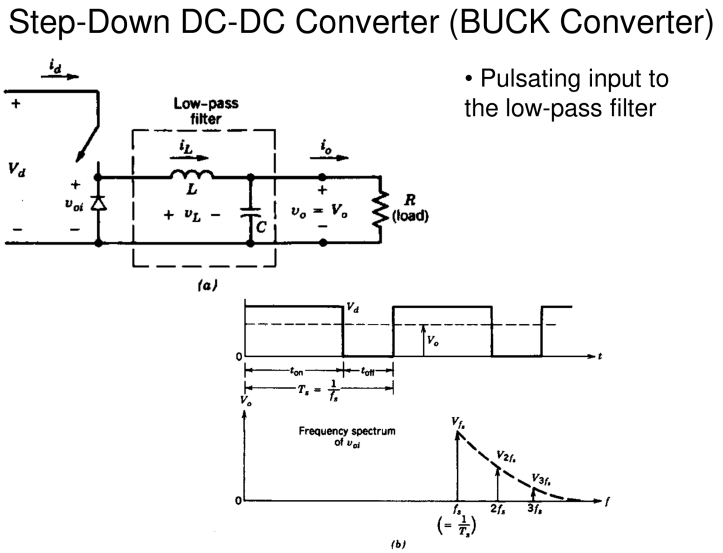

Figure 25: Step-Down (Buck) Converter: Basic circuit topology consisting of power switch S, freewheeling diode D, and output low-pass filter $L - C$.

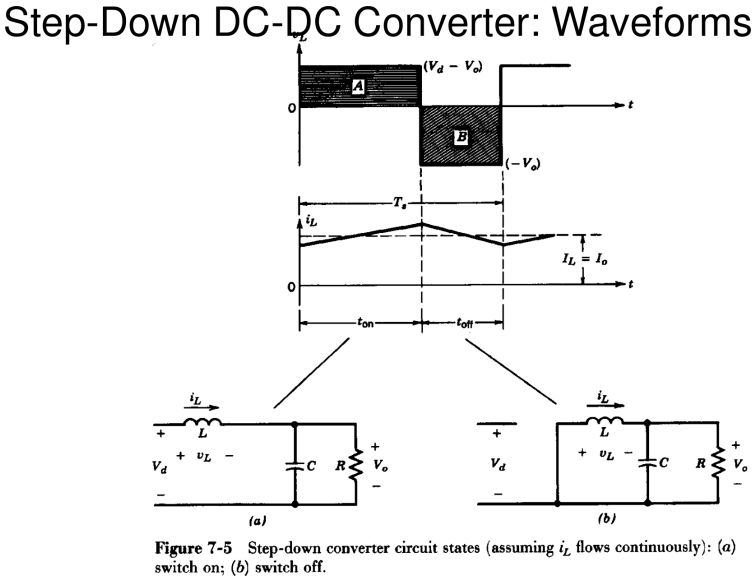

Figure 7-5 Step-down converter circuit states (assuming i_L flows continuously): (*a*) switch on; (*b*) switch off.

Figure 26: Buck Converter CCM Waveforms: Switch state, diode voltage v_D, inductor voltage v_L, triangular inductor current i_L, capacitor current i_C, and AC ripple voltage Δv_o.

Circuit Operation & Sub-Interval Analysis

The Buck converter switches between two circuit states at high frequency $f_s = 1/T_s$:

- **State 1: Switch ON ($0 \leq t \leq DT_s$):** Switch S is closed; diode D is reverse-biased by input voltage V_d. The inductor is connected between input V_d and output V_o. Applying KVL:

$$v_L(t) = V_d - V_o > 0$$

Because $v_L > 0$, the inductor current ramps **upward** linearly with constant positive slope:

$$\frac{di_L}{dt} = \frac{V_d - V_o}{L}$$

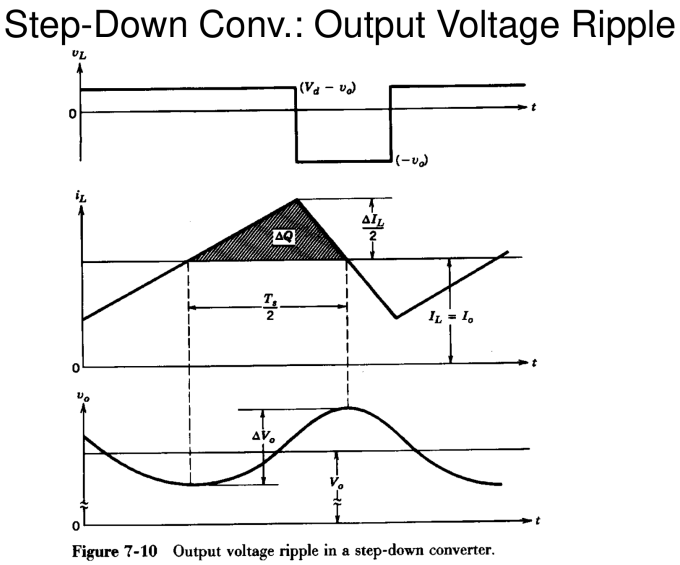

Figure 7-10 Output voltage ripple in a step-down converter.

Figure 27: Buck Converter Output Voltage Ripple: Geometric charge integration of the positive shaded triangular area $\Delta Q = \frac{T_s \Delta I_L}{8}$.

During this time, the input DC source simultaneously feeds the load and stores magnetic energy in the inductor.

- **State 2: Switch OFF ($DT_s \leq t \leq T_s$):** Switch S is opened. The collapsing magnetic field of the inductor forces current to continue flowing in the same direction, which forward-biases freewheeling diode D. The diode clamps the left side of the inductor to 0 V (ground). Applying KVL:

$$v_L(t) = 0 - V_o = -V_o < 0$$

Because $v_L < 0$, the inductor current ramps **downward** linearly with constant negative slope:

$$\frac{di_L}{dt} = -\frac{V_o}{L}$$

During this interval, the inductor discharges its stored magnetic energy into the load capacitor circuit.

Derivation of Voltage Gain (Volt-Second Balance)

In steady state, the average voltage across an ideal inductor over one complete switching period T_s must be exactly zero:

$$\int_0^{T_s} v_L(t)\,dt = \underbrace{(V_d - V_o)(DT_s)}_{\text{Volt-seconds during ON}} + \underbrace{(-V_o)((1-D)T_s)}_{\text{Volt-seconds during OFF}} = 0$$

Dividing through by T_s:

$$(V_d - V_o)D - V_o(1-D) = 0 \implies V_dD - V_oD - V_o + V_oD = 0$$

Buck Converter Voltage Conversion Ratio

$$V_o = DV_d$$

Since the duty cycle is constrained to $0 < D < 1$, the output voltage is always strictly less than the input voltage ($V_o < V_d$). Hence the name **Step-Down (Buck)**.

Inductor Ripple & Critical Inductance $L_{\min}$ for CCM

The peak-to-peak inductor current ripple ΔI_L is the net rise in current during the ON interval:

$$\Delta I_L = \left(\frac{di_L}{dt}\right) \times (DT_s) = \frac{V_d - V_o}{L} DT_s = \frac{V_d(1-D)}{L} DT_s = \frac{V_o(1-D)}{f_s L}$$

The average inductor current equals the DC load current: $I_L = I_o = \frac{V_o}{R}$.

For **Continuous Conduction Mode (CCM)**, the inductor current must never touch zero. This requires the valley current to be positive:

$$I_{L,\min} = I_L - \frac{\Delta I_L}{2} \geq 0 \implies I_L \geq \frac{\Delta I_L}{2}$$

At the boundary between CCM and Discontinuous Conduction Mode (DCM):

$$\frac{V_o}{R} = \frac{1}{2}\left[\frac{V_o(1-D)}{f_s L_{\min}}\right] \implies L_{\min} = \frac{(1-D)R}{2f_s}$$

If $L < L_{\min}$, the inductor current drops to zero before the end of the period (DCM), causing the output voltage to rise above DV_d and become load-dependent.

Output Voltage Ripple & Capacitor Sizing $C_{\min}$

Refer to Figure 24 showing the capacitor current $i_C(t) = i_L(t) - I_o$. The capacitor absorbs current whenever $i_L(t) > I_o$ (the upper shaded triangle of height $\Delta I_L/2$ and base width $T_s/2$). The total charge accumulated during this half-cycle is the area of that triangle:

$$\Delta Q = \frac{1}{2} \times \left(\frac{T_s}{2}\right) \times \left(\frac{\Delta I_L}{2}\right) = \frac{T_s \Delta I_L}{8} = \frac{\Delta I_L}{8 f_s}$$

Since $\Delta V_o = \frac{\Delta Q}{C}$:

$$\Delta V_o = \frac{\Delta I_L}{8 C f_s} = \frac{\frac{V_o(1-D)}{f_s L}}{8 C f_s} = \frac{V_o(1-D)}{8 L C f_s^2}$$

Buck Capacitor Sizing Equation

$$\frac{\Delta V_o}{V_o} = \frac{1-D}{8 L C f_s^2} \implies C_{\min} = \frac{1-D}{8 L f_s^2 (\Delta V_o / V_o)}$$

Notice the f_s^2 term in the denominator: doubling the switching frequency reduces the required filter capacitance by a factor of 4!

Step-Up (Boost) Converter

Step-Up DC-DC Converter (Boost Converter)

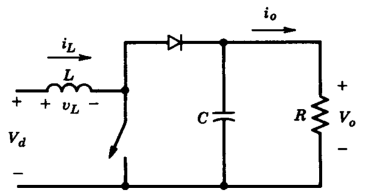

Figure 7-11 Step-up dc–dc converter.

• Output voltage must be greater than the input

Figure 28: Step-Up (Boost) Converter: Circuit topology showing inductor in series with the input DC source.

Continuous current conduction mode

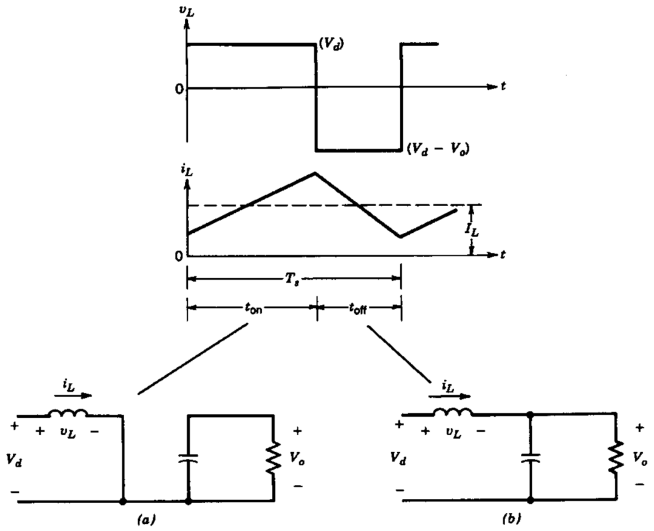

Figure 7-12 Continuous-conduction mode: (*a*) switch on; (*b*) switch off.

Figure 29: Boost Converter CCM Waveforms: Switch control signal, inductor voltage v_L, continuous inductor current i_L, discontinuous diode current i_D, and capacitor current i_C.

How Does the Voltage Get Stepped Up?

A beginner often wonders: *"How can a passive inductor output a higher DC voltage than the input battery?"* The key lies in the two-stage energy transfer:

- **Stage 1: Charging the Inductor ($0 \leq t \leq DT_s$):** Switch S is closed. The diode D is reverse-biased because cathode is at $V_o > 0$. The inductor is tied directly across input voltage V_d:

$$v_L(t) = V_d > 0 \implies \frac{di_L}{dt} = \frac{V_d}{L}$$

  Energy is drawn from the input source and stored in the inductor's magnetic field. Meanwhile, the output capacitor C alone supplies the entire load current I_o.

- **Stage 2: Boosting & Discharging ($DT_s \leq t \leq T_s$):** Switch S opens. Inductor current cannot drop instantly! The collapsing magnetic field causes the inductor terminal voltage to flip polarity (left side positive, right side higher

positive). The inductor now acts as an **auxiliary booster battery connected in series with the input source**:

$$v_L(t) = V_d - V_o < 0$$

Because the switch terminal reaches V_o, the diode conducts, dumping current from *both* the input source and the discharging inductor simultaneously into the output capacitor and load. The resulting output voltage is necessarily higher than the input: $V_o > V_d$!

Volt-Second Balance & Derivations

Applying Inductor Volt-Second Balance:

$$\int_0^{T_s} v_L(t)\,dt = (V_d)(DT_s) + (V_d - V_o)((1-D)T_s) = 0$$

$$V_d D + (V_d - V_o)(1-D) = 0 \implies V_d[D + 1 - D] = V_o(1-D)$$

Boost Converter Voltage Gain

$$V_o = \frac{V_d}{1-D}$$

Because $0 < D < 1$, the factor $\frac{1}{1-D} > 1$, meaning $V_o > V_d$ for all duty cycles.

Inductor Current & Sizing Equations

Assuming an ideal converter (100% efficiency), input power equals output power:

$$P_{\text{in}} = P_{\text{out}} \implies V_d I_{\text{in}} = V_o I_o$$

Since the input source is in direct series with the inductor, the average input current is the average inductor current: $I_{\text{in}} = I_L$. Therefore:

$$I_L = \frac{V_o}{V_d} I_o = \frac{1}{1-D} I_o = \frac{V_d}{(1-D)^2 R}$$

- Inductor Current Ripple: $\Delta I_L = \frac{V_d D T_s}{L} = \frac{V_d D}{f_s L}$
- Boundary Condition for CCM ($I_L \geq \Delta I_L/2$):

$$\frac{V_d}{(1-D)^2 R} \geq \frac{1}{2}\left[\frac{V_d D}{f_s L_{\text{min}}}\right] \implies L_{\text{min}} = \frac{D(1-D)^2 R}{2 f_s}$$

- Capacitor Ripple & Sizing: During the switch ON interval (DT_s), the diode is OFF, so the capacitor must supply the full load current $I_o = V_o/R$. The total charge removed from C is $\Delta Q = I_o(DT_s)$. Therefore:

$$\Delta V_o = \frac{\Delta Q}{C} = \frac{I_o D T_s}{C} = \frac{V_o D}{RC f_s} \implies \frac{\Delta V_o}{V_o} = \frac{D}{RC f_s} \implies C_{\text{min}} = \frac{D}{R f_s (\Delta V_o / V_o)}$$

Step-Down/Step-Up (Buck-Boost) Converter

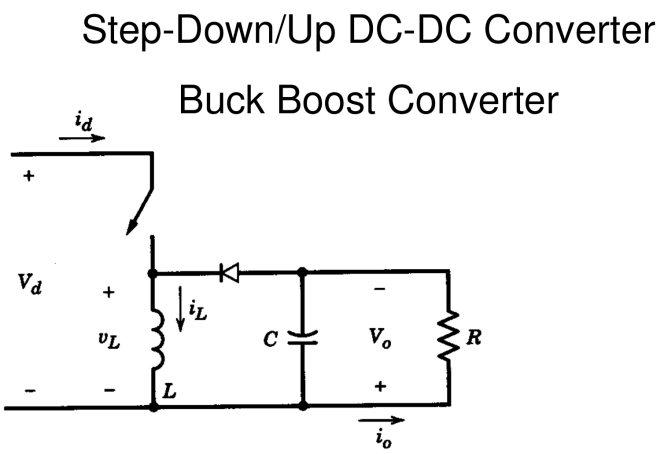

Figure 30: Step-Down/Step-Up (Buck-Boost) Converter: Circuit schematic showing reversed diode polarity yielding negative output voltage.

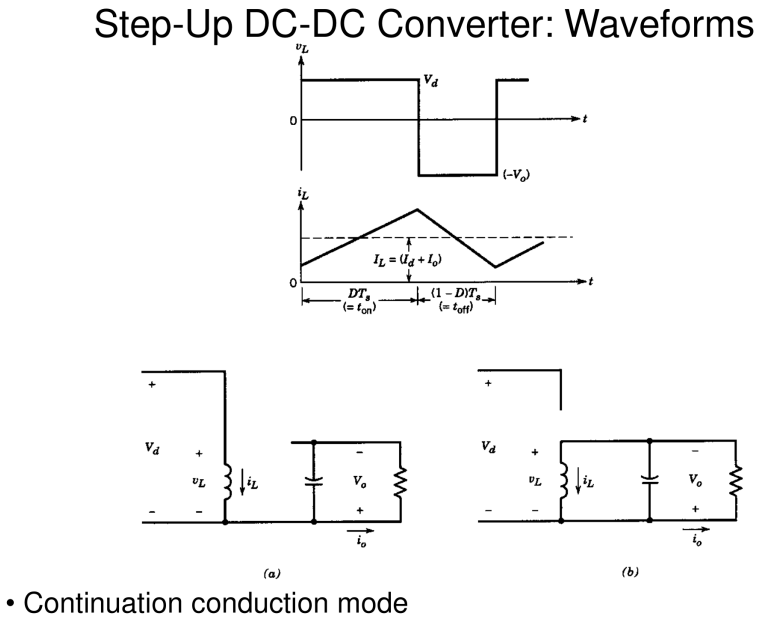

Figure 31: Buck-Boost Converter CCM Waveforms: Switch signal, inductor voltage v_L, inductor current i_L, diode current, and capacitor current.

Circuit Operation & Why Is Output Polarity Inverted?

Look closely at the schematic in Figure 27: the diode cathode is connected to the inductor, and its anode is connected to the output DC rail!

- **Switch ON ($0 \leq t \leq DT_s$):** Switch S is closed. The diode D is reverse-biased. The inductor connects directly across the input source V_d, drawing current from top to bottom:

$$v_L(t) = V_d > 0 \implies \frac{di_L}{dt} = \frac{V_d}{L}$$

  Energy is stored entirely in the inductor. The load is completely isolated from the source and is powered solely by capacitor C.

- **Switch OFF ($DT_s \leq t \leq T_s$):** Switch S opens. The inductor current **must continue flowing in the same downward direction** (into ground). To sustain this downward flow, current is pulled out of the bottom ground rail and pushed *upward* into the anode of diode D!

  This forces the top rail of the load to become **negative with respect to ground**. Hence, the output voltage is inverted:

$$v_L(t) = V_o < 0 \quad \text{(negative relative to ground)}$$

Volt-Second Balance & Sizing Formulae

Applying Inductor Volt-Second Balance:

$$(V_d)(DT_s) + (V_o)((1-D)T_s) = 0 \implies V_o = -\frac{D}{1-D}V_d \implies |V_o| = \frac{D}{1-D}V_d$$

Buck-Boost Operational Regimes

- For $D < 0.5 \implies \frac{D}{1-D} < 1 \implies |V_o| < V_d$ **(Step-Down / Buck Mode)**.
- For $D = 0.5 \implies \frac{D}{1-D} = 1 \implies |V_o| = V_d$ (Unity gain inverter).
- For $D > 0.5 \implies \frac{D}{1-D} > 1 \implies |V_o| > V_d$ **(Step-Up / Boost Mode)**.

Component Sizing Equations

- Average Inductor Current: $I_L = I_{\text{in}} + I_o = \frac{D}{1-D}I_o + I_o = \frac{I_o}{1-D} = \frac{DV_d}{(1-D)^2R}$
- Inductor Current Ripple: $\Delta I_L = \frac{V_d D}{f_s L}$
- Critical Inductance for CCM ($I_L \geq \Delta I_L/2$):

$$\frac{|V_o|}{(1-D)R} \geq \frac{1}{2}\left[\frac{V_d D}{f_s L_{\text{min}}}\right] \implies L_{\text{min}} = \frac{(1-D)^2 R}{2f_s}$$

- Output Voltage Ripple: During switch ON, diode is OFF and capacitor supplies the full load current I_o:

$$\Delta Q = I_o(DT_s) \implies \frac{\Delta V_o}{|V_o|} = \frac{D}{RCf_s} \implies C_{\text{min}} = \frac{D}{Rf_s(\Delta V_o/|V_o|)}$$

Effect of Non-Idealities

Inductor Resistance r_L in Boost Converter

Due to inductor winding resistance r_L, the actual voltage gain of a Boost converter is:

$$\frac{V_o}{V_d} = \frac{1}{1-D} \cdot \frac{1}{1 + \frac{r_L}{R(1-D)^2}}$$

As $D \to 1$, the inductor current $I_L = I_o/(1-D)$ becomes massive, so $I_L^2 r_L$ losses completely dominate! The voltage gain peaks around $D \approx 0.85$ and drops sharply to zero at $D = 1$.

Capacitor ESR (r_C)

Capacitor Equivalent Series Resistance produces an additional voltage step: $\Delta V_{\text{ESR}} = \Delta I \cdot r_C$. At switching frequencies above 50 kHz, ESR ripple often exceeds pure capacitive ripple!

Selection of Power Semiconductor Devices

Device Selection Guidelines for DC-DC Converters

- **Power MOSFET:** Preferred for low-to-medium voltage (< 500 V), high-frequency (50 kHz to several MHz) applications. Very fast switching transitions (< 100 ns) minimise switching losses. Conduction loss is $I_D^2 \cdot R_{DS(on)}$, which increases with voltage rating. Ideal choice for **Buck converters** at $V_d < 200$ V.
- **IGBT (Insulated Gate Bipolar Transistor):** Preferred for medium-to-high voltage (> 600 V), moderate-frequency (< 50 kHz) applications. Lower conduction loss than MOSFET at high voltages due to bipolar conductivity modulation. Limited by *current tailing* during turn-off. Ideal for **Boost/Buck-Boost converters** at higher power levels.
- **Schottky Diode:** Negligible reverse recovery time ($t_{rr} \approx 0$) and low forward voltage drop ($V_F \approx 0.3$–0.5 V). Ideal freewheeling diode in low-voltage SMPS ($V_o < 100$ V) where recovery losses of p-n junction diodes would be unacceptable.
- **Fast Recovery Diode (FRD):** Used as freewheeling diode in higher voltage (> 200 V) converters where Schottky diodes are unavailable. Has short but non-zero t_{rr} (< 100 ns).

Key Trade-off: Higher switching frequency f_s reduces filter L and C sizes (smaller, lighter converter), but increases switching losses. The device choice must balance these factors.

Comprehensive Solved Numerical Problems

Problem 1: 1-Phase Full Converter RL Load Performance

Question: A single-phase full converter operates from $V_s = 230$ V, 50 Hz supply with $\alpha = 45°$ feeding a ripple-free current $I_o = 15$ A. Calculate V_o, $V_{o,\text{rms}}$, P, I_{s1}, DF, g, PF, and THD.

Solution:

1. Peak voltage: $V_m = \sqrt{2} \times 230 = 325.27$ V.
2. Average Output Voltage:
$$V_o = \frac{2V_m}{\pi}\cos\alpha = \frac{2 \times 325.27}{\pi}\cos(45°) = 207.07 \times 0.7071 = \mathbf{146.42}\text{ V}$$
3. RMS Output Voltage: $V_{o,\text{rms}} = V_s = \mathbf{230}$ V.
4. Real Power: $P = V_o I_o = 146.42 \times 15 = \mathbf{2196.3}$ W.
5. Fundamental RMS Current: $I_{s1} = \frac{2\sqrt{2}}{\pi} I_o = 0.9003 \times 15 = \mathbf{13.505}$ A.
6. Displacement Factor: $DF = \cos(45°) = \mathbf{0.7071}$ (lagging).
7. Distortion Factor: $g = \frac{I_{s1}}{I_s} = \frac{13.505}{15} = \mathbf{0.9003}$.
8. Input Power Factor: $PF = g \times DF = 0.9003 \times 0.7071 = \mathbf{0.6366}$ (lagging).
9. Current THD: $THD = \sqrt{\frac{1}{g^2} - 1} = \sqrt{\frac{\pi^2}{8} - 1} = \mathbf{48.34\%}$.

Problem 2: 1-Phase Full Converter Inverter Mode

Question: A 230 V, 50 Hz full converter operates in inverter mode to regenerate power into the AC grid from a 150 V battery ($R = 1.5\ \Omega$) with $I_o = 12$ A. Find α and power regenerated.

Solution:

1. Loop KVL in inverter mode:
$$V_o = -(E - I_o R) = -(150 - 12 \times 1.5) = -(150 - 18) = \mathbf{-132}\text{ V}$$
2. Since $V_o = \frac{2V_m}{\pi}\cos\alpha$:
$$-132 = 207.07\cos\alpha \implies \cos\alpha = \frac{-132}{207.07} = -0.63746$$
$$\alpha = \cos^{-1}(-0.63746) = \mathbf{129.6}°$$
3. Power delivered to AC source: $P_{\text{regen}} = |V_o| I_o = 132 \times 12 = \mathbf{1584}\text{ W} = \mathbf{1.584}\text{ kW}$.

Problem 3: Semi-Converter vs. Full Converter Power Factor

Question: A single-phase semi-converter operates from 230 V, 50 Hz at $\alpha = 60°$ with $I_o = 10$ A. Calculate V_o, I_s, DF, and PF. Compare with a full converter.

Solution:

1. $V_o = \frac{V_m}{\pi}(1+\cos 60^\circ) = \frac{325.27}{\pi}(1.5) = \mathbf{155.31}$ V.
2. $I_s = I_o\sqrt{\frac{180^\circ-60^\circ}{180^\circ}} = 10 \times \sqrt{2/3} = \mathbf{8.165}$ A.
3. $DF = \cos(60^\circ/2) = \cos(30^\circ) = \mathbf{0.8660}$.
4. $P = V_o I_o = 1553.1$ W; $\quad S = 230 \times 8.165 = 1877.95$ VA $\implies PF = \frac{1553.1}{1877.95} = \mathbf{0.8270}$.
5. Full converter at 60°: $PF_{\text{full}} = 0.9003\cos(60^\circ) = \mathbf{0.4502}$.

$$\frac{PF_{\text{semi}}}{PF_{\text{full}}} = \frac{0.8270}{0.4502} \approx \mathbf{1.84} \quad (\mathbf{84}\%\text{ higher!})$$

Problem 4: Source Inductance in 1-Phase Full Converter

Question: $V_s = 230\,\text{V}, 50\,\text{Hz}, L_s = 2.5\,\text{mH}, I_o = 20\,\text{A}, \alpha = 30°$. Find μ, ΔV_o, V_o, and R_c.

Solution:

1. $\frac{2\omega L_s I_o}{V_m} = \frac{2\times 314.16\times 0.0025\times 20}{325.27} = 0.09658.$
2. $\cos(30° + \mu) = \cos(30°) - 0.09658 = 0.86603 - 0.09658 = 0.76945.$

$$30° + \mu = \cos^{-1}(0.76945) = 39.70° \implies \mu = \mathbf{9.70}°$$

3. $\Delta V_o = 2 f L_s I_o = 2 \times 50 \times 0.0025 \times 20 = \mathbf{5.0}\,\text{V}.$
4. $V_{o,\text{ideal}} = 207.07\cos(30°) = 179.33\,\text{V} \implies V_o = 179.33 - 5.0 = \mathbf{174.33}\,\text{V}.$
5. $R_c = 2 f L_s = \mathbf{0.25}\,\Omega.$

Problem 5: 3-Phase Full Converter Feeding SEDC Motor

Question: A 3-phase full converter operates from 400 V line, 50 Hz. It feeds a DC motor with $E = 350\,\text{V}$, $R = 0.5\,\Omega$, $I_o = 40\,\text{A}$. Find α, P, and input PF.

Solution:

1. Required terminal voltage: $V_o = E + I_o R = 350 + (40 \times 0.5) = \mathbf{370}\,\text{V}.$
2. Peak line voltage: $V_{ml} = \sqrt{2} \times 400 = 565.69\,\text{V}.$

$$V_o = \frac{3V_{ml}}{\pi}\cos\alpha \implies 370 = \frac{3 \times 565.69}{\pi}\cos\alpha = 540.19\cos\alpha$$

$$\cos\alpha = \frac{370}{540.19} = 0.6849 \implies \alpha = \mathbf{46.77}°$$

3. Active Power: $P = V_o I_o = 370 \times 40 = \mathbf{14.8}\,\text{kW}.$
4. Input Power Factor: $PF = \frac{3}{\pi}\cos\alpha = 0.9549 \times 0.6849 = \mathbf{0.654}$ (lagging).

Problem 6: 3-Phase Semi-Converter

Question: Repeat Problem 5 if a 3-phase semi-converter is used.

Solution:

$$V_o = \frac{3V_{ml}}{2\pi}(1 + \cos\alpha) \implies 370 = 270.095(1 + \cos\alpha)$$

$$1 + \cos\alpha = \frac{370}{270.095} = 1.36989 \implies \cos\alpha = 0.36989 \implies \alpha = \mathbf{68.29}°$$

Since $\alpha > 60°$, it operates in the freewheeling 3-pulse regime.

Problem 7: 3-Phase Full Converter with Source Inductance

Question: $V_L = 415\,\text{V}, 50\,\text{Hz}, L_s = 1.2\,\text{mH}, I_o = 50\,\text{A}, \alpha = 45^\circ$. Find μ, ΔV_o, and V_o.

Solution:

1. $V_{ml} = \sqrt{2} \times 415 = 586.9\,\text{V}$.

$$\cos\alpha - \cos(\alpha+\mu) = \frac{2\omega L_s I_o}{\sqrt{3}V_{ml}} = \frac{2 \times 314.16 \times 0.0012 \times 50}{\sqrt{3} \times 586.9} = 0.037085$$

$$\cos(45^\circ + \mu) = \cos(45^\circ) - 0.037085 = 0.70711 - 0.037085 = 0.67002$$

$$45^\circ + \mu = 47.93^\circ \implies \mu = \mathbf{2.93}^\circ$$

2. Voltage Drop: $\Delta V_o = 3 f L_s I_o = 3 \times 50 \times 0.0012 \times 50 = \mathbf{9.0}\,\text{V}$.

3. Ideal: $V_{o,\text{ideal}} = 560.46\cos(45^\circ) = 396.31\,\text{V} \implies V_o = 396.31 - 9.0 = \mathbf{387.31}\,\text{V}$.

Problem 8: Complete Buck Converter Design

Question: $V_d = 48\,\text{V}, V_o = 12\,\text{V}, R = 6\,\Omega, f_s = 40\,\text{kHz}$. Find D, L_{min}, actual ripples for $L = 1.5L_{\text{min}}$, and C for $\Delta V_o/V_o \le 0.5\%$.

Solution:

1. $D = \frac{V_o}{V_d} = \frac{12}{48} = \mathbf{0.25}$.

2. $L_{\text{min}} = \frac{(1-D)R}{2f_s} = \frac{(1-0.25)\times 6}{2\times 40{,}000} = \mathbf{56.25}\,\mu\text{H}$.

3. $L = 1.5 \times 56.25 = 84.375\,\mu\text{H}$.

$$\Delta I_L = \frac{V_o(1-D)}{f_s L} = \frac{12 \times 0.75}{40{,}000 \times 84.375 \times 10^{-6}} = \mathbf{2.667}\,\text{A}$$

$$I_{L,\text{max}} = I_o + \frac{\Delta I_L}{2} = 2 + 1.333 = \mathbf{3.333}\,\text{A}$$

4. Filter Capacitor:

$$C_{\text{min}} = \frac{1-D}{8Lf_s^2(\Delta V_o/V_o)} = \frac{0.75}{8 \times 84.375 \times 10^{-6} \times (40{,}000)^2 \times 0.005} = \mathbf{138.9}\,\mu\text{F}$$

Problem 9: Boost Converter with Inductor Resistance r_L

Question: $V_d = 12\,\text{V}, R = 50\,\Omega, D = 0.8, r_L = 0.5\,\Omega$. Find ideal V_o, actual V_o, and efficiency η.

Solution:

1. Ideal: $V_{o,\text{ideal}} = \frac{V_d}{1-D} = \frac{12}{0.2} = \mathbf{60}\,\text{V}$.

2. Parasitic factor: $\frac{r_L}{R(1-D)^2} = \frac{0.5}{50\times(0.2)^2} = \frac{0.5}{2.0} = 0.25$.

$$V_o = \frac{V_d}{1-D}\frac{1}{1+0.25} = 60 \times \frac{1}{1.25} = \mathbf{48}\,\text{V}$$

3. Efficiency: $\eta = \frac{1}{1+0.25} = \frac{1}{1.25} = \mathbf{80}\%$.

Problem 10: Buck-Boost Converter Sizing & Device Stress

Question: $V_d = 24\,\text{V}$, $|V_o| = 36\,\text{V}$, $R = 18\,\Omega$, $f_s = 50\,\text{kHz}$. Find D, I_L, L_{min}, and switch voltage stress.

Solution:

1. $\frac{|V_o|}{V_d} = \frac{D}{1-D} \implies 1.5 = \frac{D}{1-D} \implies D = \mathbf{0.60}$.
2. Output current: $I_o = \frac{36}{18} = 2\,\text{A} \implies I_L = \frac{I_o}{1-D} = \frac{2}{0.4} = \mathbf{5}\,\text{A}$.
3. $L_{\text{min}} = \frac{(1-D)^2 R}{2 f_s} = \frac{(0.4)^2 \times 18}{2 \times 50{,}000} = \mathbf{28.8}\,\mu\text{H}$.
4. Switch and Diode Peak Blocking Voltage:

$$V_{\text{switch,peak}} = V_{\text{diode,PIV}} = V_d + |V_o| = 24 + 36 = \mathbf{60}\,\text{V}$$

Master Formula Reference Summary

Table 2: Master Formula Sheet — Phase Controlled Rectifiers

Converter Topology	Average Voltage V_o	RMS Voltage $V_{o,\text{rms}}$	Displ. Factor (DF)	Input Power Factor (PF)
1-ϕ Half-Wave (R)	$\frac{V_m}{2\pi}(1+\cos\alpha)$	$\frac{V_m}{2}\sqrt{\frac{(\pi-\alpha)+\frac{1}{2}\sin 2\alpha}{\pi}}$	—	—
1-ϕ Full Converter	$\frac{2V_m}{\pi}\cos\alpha$	$V_s = \frac{V_m}{\sqrt{2}}$	$\cos\alpha$	$\frac{2\sqrt{2}}{\pi}\cos\alpha \approx 0.90\cos\alpha$
1-ϕ Semi-Converter	$\frac{V_m}{\pi}(1+\cos\alpha)$	$\frac{V_m}{\sqrt{2}}\sqrt{\frac{(\pi-\alpha)+\frac{1}{2}\sin 2\alpha}{\pi}}$	$\cos\left(\frac{\alpha}{2}\right)$	$\frac{\sqrt{2}(1+\cos\alpha)}{\sqrt{\pi(\pi-\alpha)}}$
1-ϕ Full (L_s)	$\frac{2V_m}{\pi}\cos\alpha - 2fL_sI_o$	—	—	$\Delta V_o = 2fL_sI_o$
3-ϕ 6-Pulse Bridge	$\frac{3V_{ml}}{\pi}\cos\alpha$	$V_{ml}\sqrt{\frac{1}{2}+\frac{3\sqrt{3}}{4\pi}\cos 2\alpha}$	$\cos\alpha$	$\frac{3}{\pi}\cos\alpha \approx 0.955\cos\alpha$
3-ϕ Semi-Converter	$\frac{3V_{ml}}{2\pi}(1+\cos\alpha)$	—	—	Higher than 6-pulse
3-ϕ Full (L_s)	$\frac{3V_{ml}}{\pi}\cos\alpha - 3fL_sI_o$	—	—	$\Delta V_o = 3fL_sI_o$

Table 3: Master Formula Sheet — DC-DC Switching Regulators (CCM)

Regulator Topology	Voltage Gain $\frac{V_o}{V_d}$	Inductor Current I_L	Boundary Inductance L_{min}	Capacitor Ripple $\frac{\Delta V_o}{V_o}$
Buck (Step-Down)	D	$I_o = \frac{V_o}{R}$	$\frac{(1-D)R}{2f_s}$	$\frac{1-D}{8LCf_s^2}$
Boost (Step-Up)	$\frac{1}{1-D}$	$\frac{I_o}{1-D} = \frac{V_d}{(1-D)^2R}$	$\frac{D(1-D)^2R}{2f_s}$	$\frac{D}{RCf_s}$
Buck-Boost	$-\frac{D}{1-D}$	$\frac{I_o}{1-D} = \frac{DV_d}{(1-D)^2R}$	$\frac{(1-D)^2R}{2f_s}$	$\frac{D}{RCf_s}$

— End of Module 2 Master Study Guide —